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Disclaimer

This utility is not intended for use in a production setting – security was not a consideration of the design. If you are operating a production system, use a commercial keyloader that supports P25 link layer authentication.

The subscriber identity and the link layer authentication key loaded into to the radio need to be entered into the trunking system in order for the authentication process to be successful. This tool by itself will not allow access to a link layer authentication protected trunking system.

Version History

Version #	Date	Description
1.0.0.0	2019-01-17	Initial Release

Operation

Overview

This utility is a Windows .NET Console Application that implements all Manual Rekeying Features for Authentication per TIA-102.AACD-A.

The Microsoft .NET Framework 4.6.1 (or later) is required to be installed.

The following command line flags are available:

-h, -?, --help	show help and exit
-q, --quiet	do not show output
-v, --verbose	show debug messages
-i, --ip=VALUE	radio ip address [default 192.168.128.1]
-p, --port=VALUE	radio udp port number [default 49165]
-t, --timeout=VALUE	radio receive timeout (ms) [default 5000]
-l, --load	load key
-z, --zeroize	zeroize key(s)
-r, --read	read key(s)
-d, --device	device scope
-a, --active	active scope
-n, --named	named scope
-w, --wacn=VALUE	wacn id (hex)
-s, --system=VALUE	system id (hex)
-u, --unit=VALUE	unit id (hex)
-k, --key=VALUE	aes-128 encryption key (hex)

The /wacn, /system, and /unit, and /key flags all expect hex values.

If the load action is specified, the key is required.

If the named scope is specified, the WACN ID, System ID, and Unit ID are required.

Radio IP 192.168.128.1 and UDP port 49165 are the default. This combination is used by Motorola radios, however other vendors may be different. Connection over PPP (through the RS-232 serial driver) or over RNDIS is supported, as that configuration is on the Windows level.

Load

This action loads a key into the radio by sending a 'Load Authentication Key Command' KMM.

Only the active and named scopes are supported.

Example: Load the link layer authentication key 0x000102030405060708090A0B0C0D0E0F into the active subscriber identity.

sakl.exe /load /active /key 000102030405060708090a0b0c0d0e0f

Example: Load the link layer authentication key 0x000102030405060708090A0B0C0D0E0F into the subscriber identity defined by WACN ID: 0xA4398, System ID: 0xF10, Unit ID: 0x99B584.

sakl.exe /load /named /wacn a4398 /system f10 /unit 99b584 /key 000102030405060708090a0b0c0d0e0f

This action returns 0 if the operation completed successfully, or -1 if there was an error during the operation.

Zeroize

This action deletes key(s) in the radio by sending a 'Delete Authentication Key Command' KMM.

Example: Delete all link layer authentication keys in the device.

sakl.exe /zeroize /device

Example: Delete the link layer authentication key associated with the active subscriber identity.

sakl.exe /zeroize /active

Example: Delete the link layer authentication key associated with the subscriber identity defined by WACN ID: 0xA4398, System ID: 0xF10, Unit ID: 0x99B584.

sakl.exe /zeroize /named /wacn a4398 /system f10 /unit 99b584

This action returns 0 if the operation completed successfully, or -1 if there was an error during the operation.

Read

This action lists the status of the key(s) in the radio by sending an 'Inventory Command (List Active SUID)' KMM.

Only the device and active scopes are supported.

NOTE: This does not read the encryption key itself, but the status of the key.

Example: List all link layer authentication keys in the device.

sakl.exe /read /device

Example: List the link layer authentication key associated with the active subscriber identity.

sakl.exe /read /active

The following information is printed for each key returned by the device:

WACN: 0xA4398, System: 0xF10, Unit: 0x99B584, Key Assigned: True, Is Active: True

This action returns 0 if the operation completed successfully, or -1 if there was an error during the operation.

RS-232 Serial Driver

NOTE: The driver connects 9600 bits per second, 8 data bits, no parity bit, 1 stop bit (9600-8-N-1) by default. In order to change baud rates, the DCB entry in the INF file has to be changed.

IMPORTANT: Start with the radio disconnected from the serial port. This prevents Windows from automatically starting the connection before the configuration is complete.

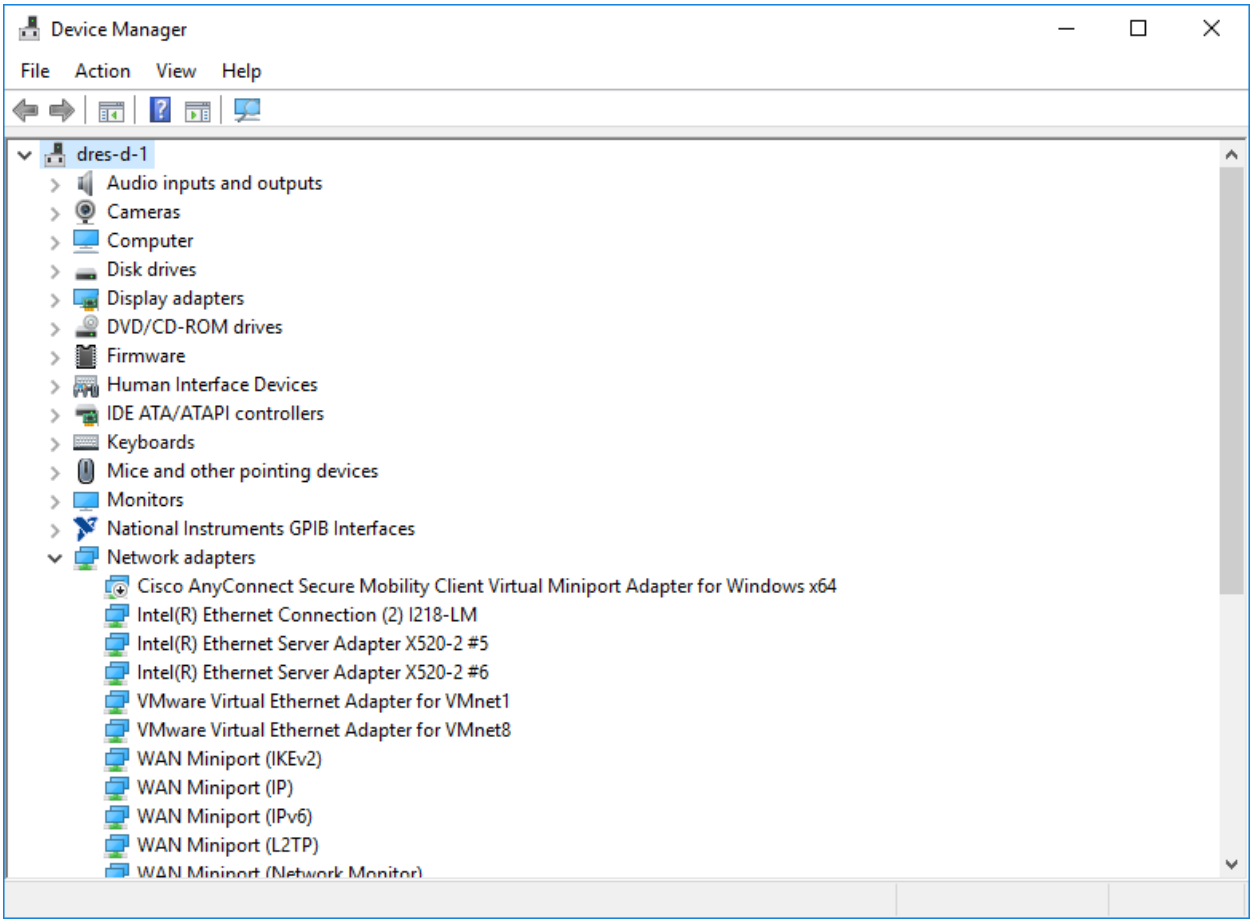
Windows 10

NOTE: The following depicts Windows 10 Enterprise, 64-bit, Build 1803. Steps on earlier or later builds may differ.

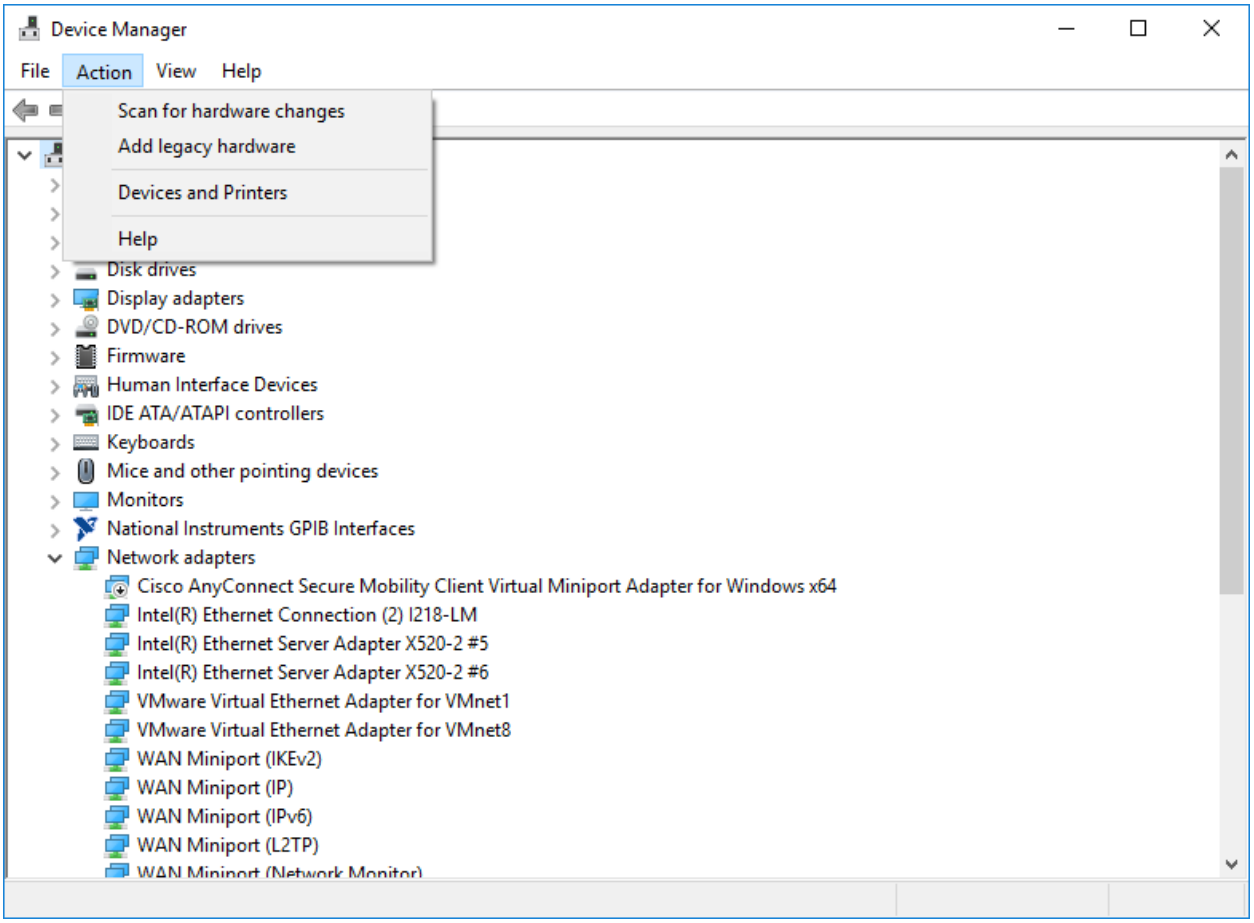
Installation

Open Device Manager.

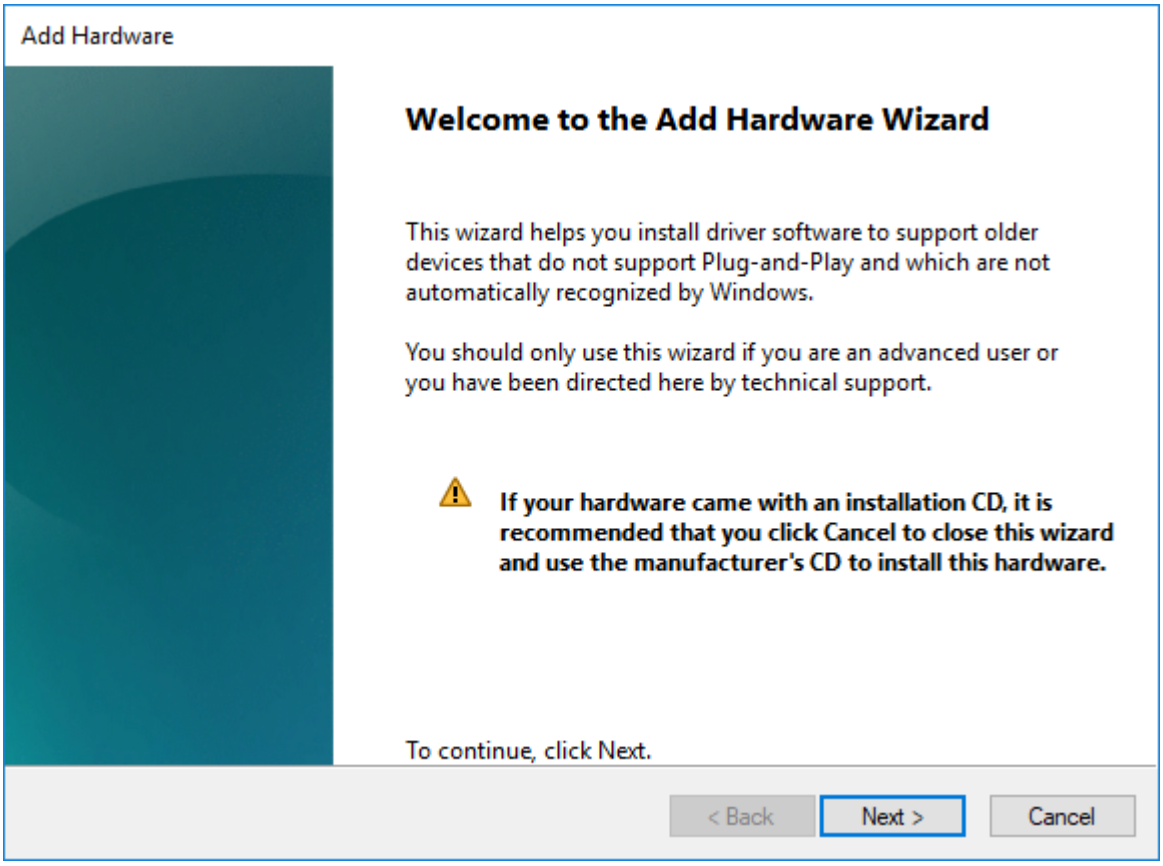
Click on the computer name.



Click the ‘Action’ menu, then ‘Add legacy hardware’.

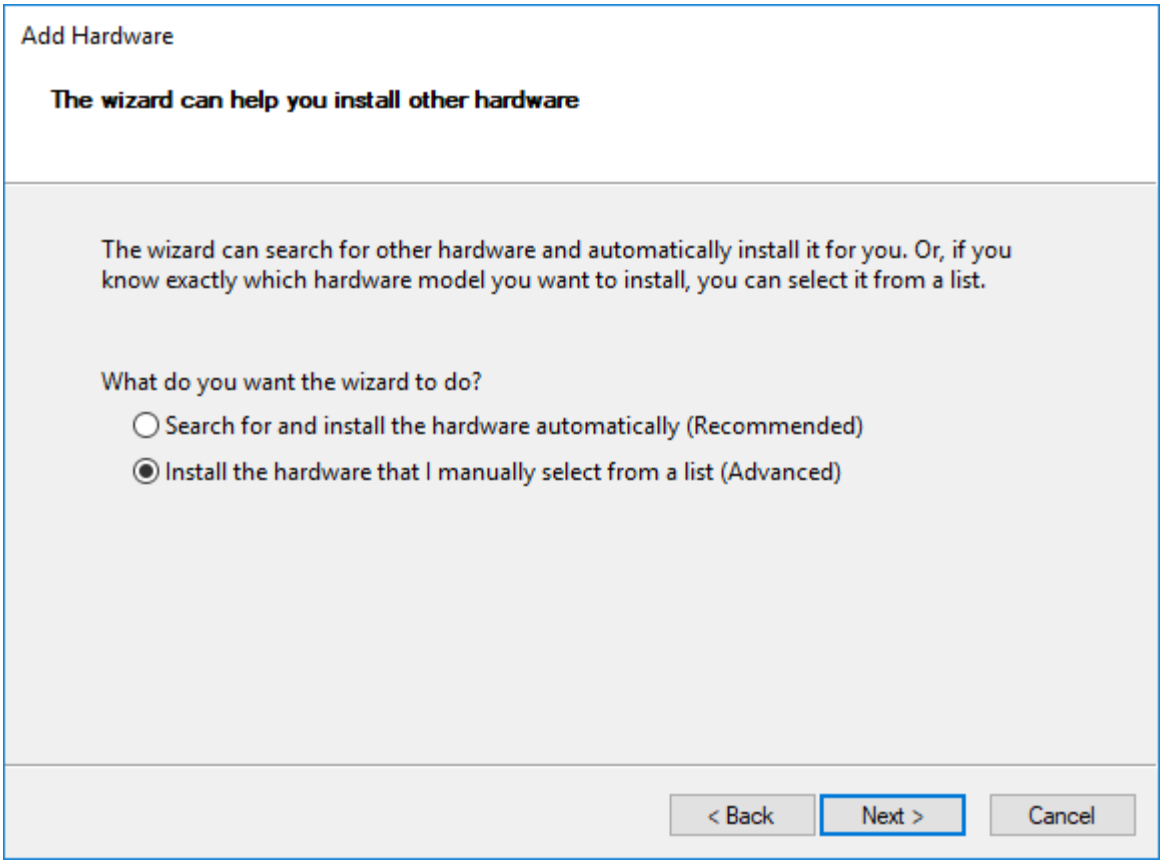


Click ‘Next’.


Add Hardware
Welcome to the Add Hardware Wizard
This wizard helps you install driver software to support older devices that do not support Plug-and-Play and which are not automatically recognized by Windows.
You should only use this wizard if you are an advanced user or you have been directed here by technical support.
If your hardware came with an installation CD, it is recommended that you click Cancel to close this wizard and use the manufacturer's CD to install this hardware.
To continue, click Next.
< Back
Next >
Cancel

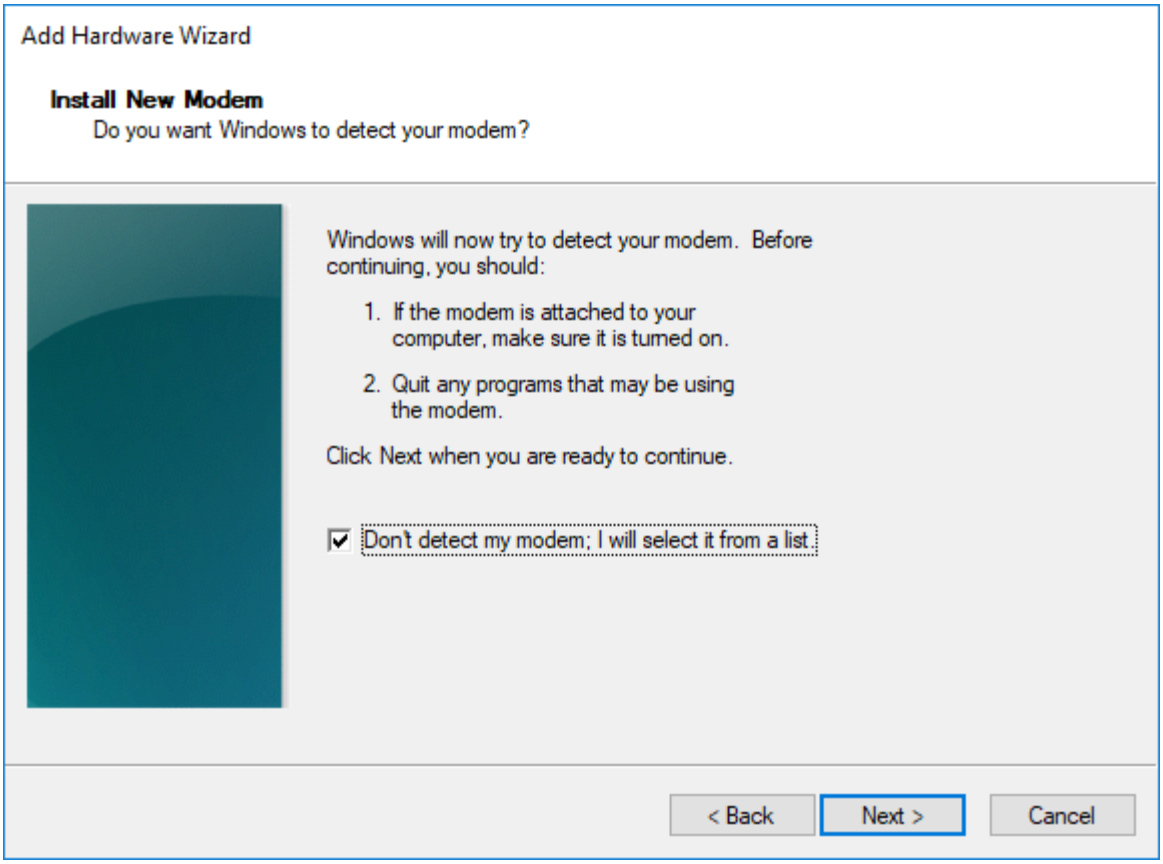
Select ‘Install the hardware that I manually select from a list (Advanced)’.

Click ‘Next’.

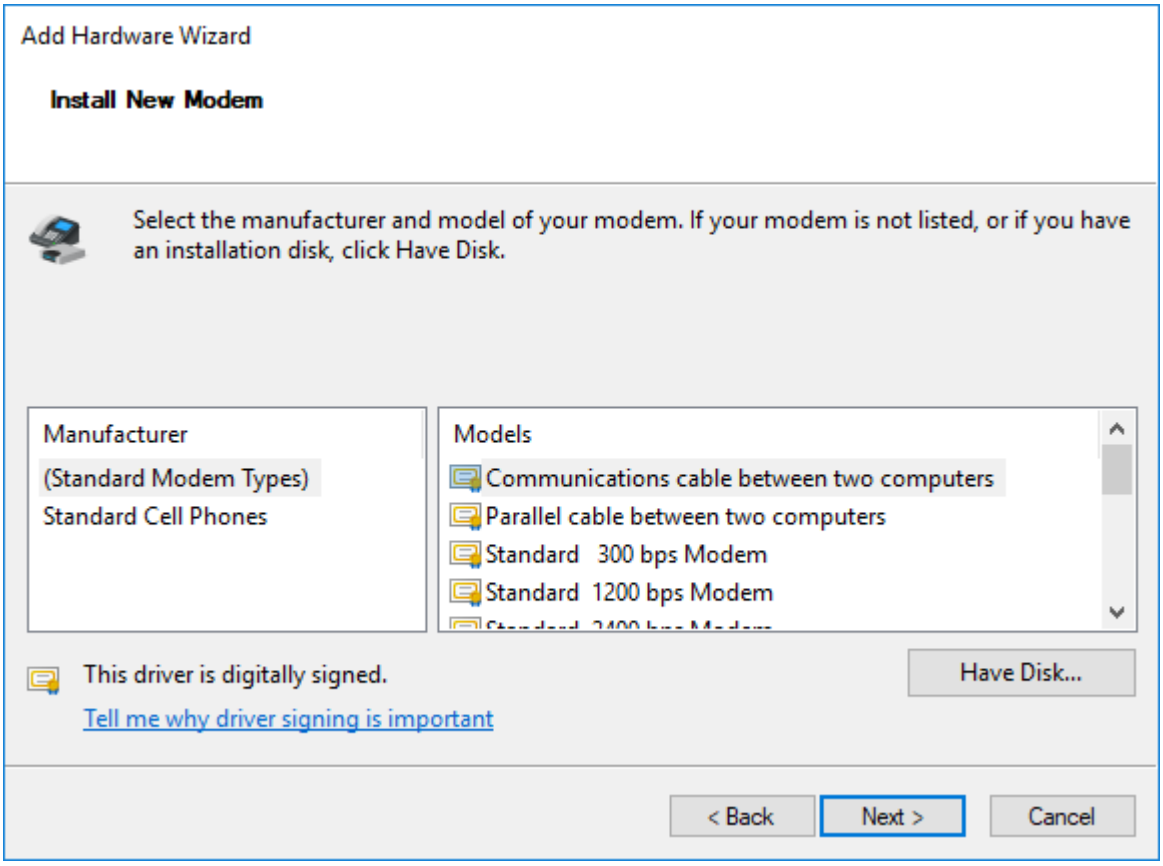


Select the ‘Don’t detect my modem; I will select it from a list’ option.

Click ‘Next’.

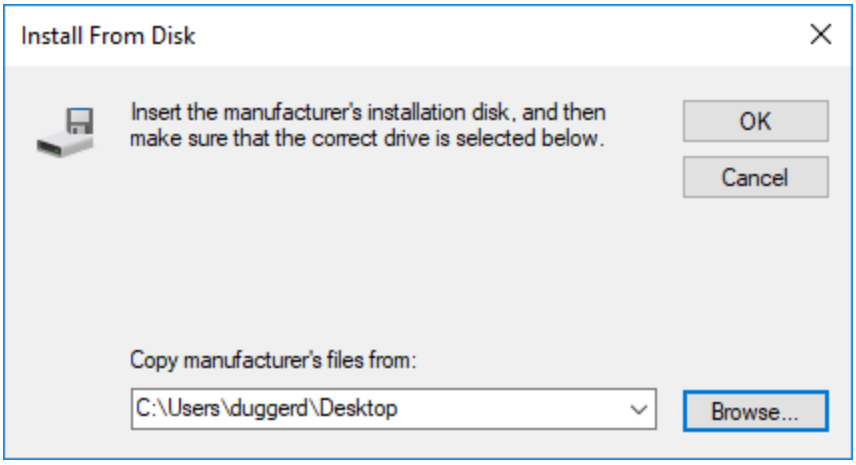


Click ‘Have Disk’.



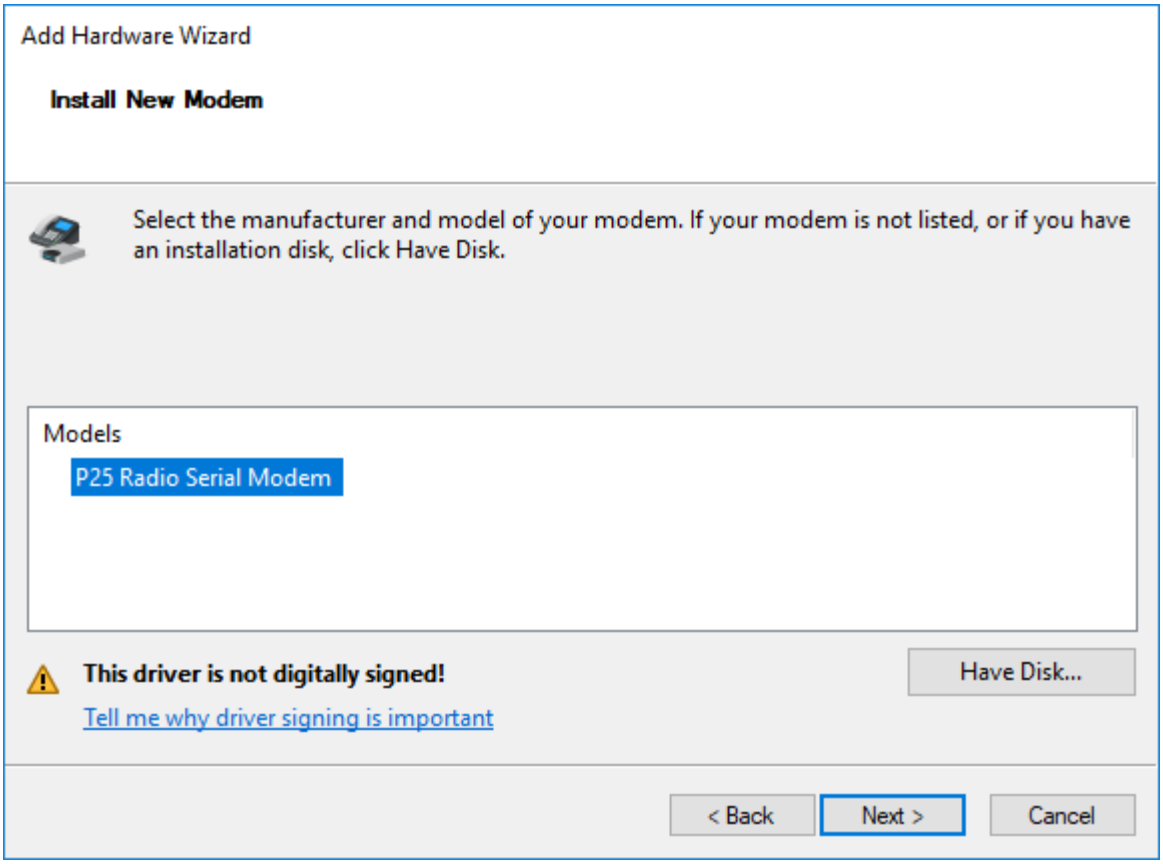
Click ‘Browse’ and navigate to the folder that contains the ‘P25RadioSerialModem.inf’ file.

Click ‘OK’.



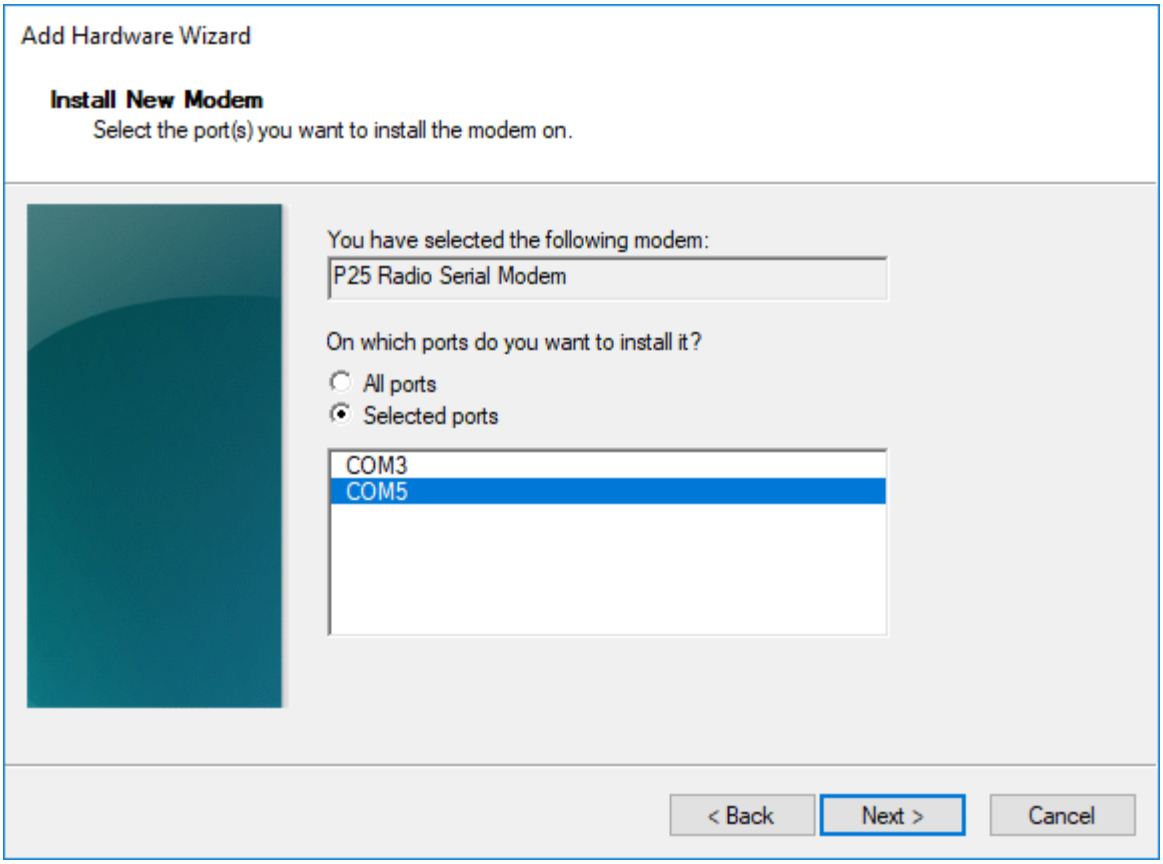
Select ‘P25 Radio Serial Modem’.

Click ‘Next’.

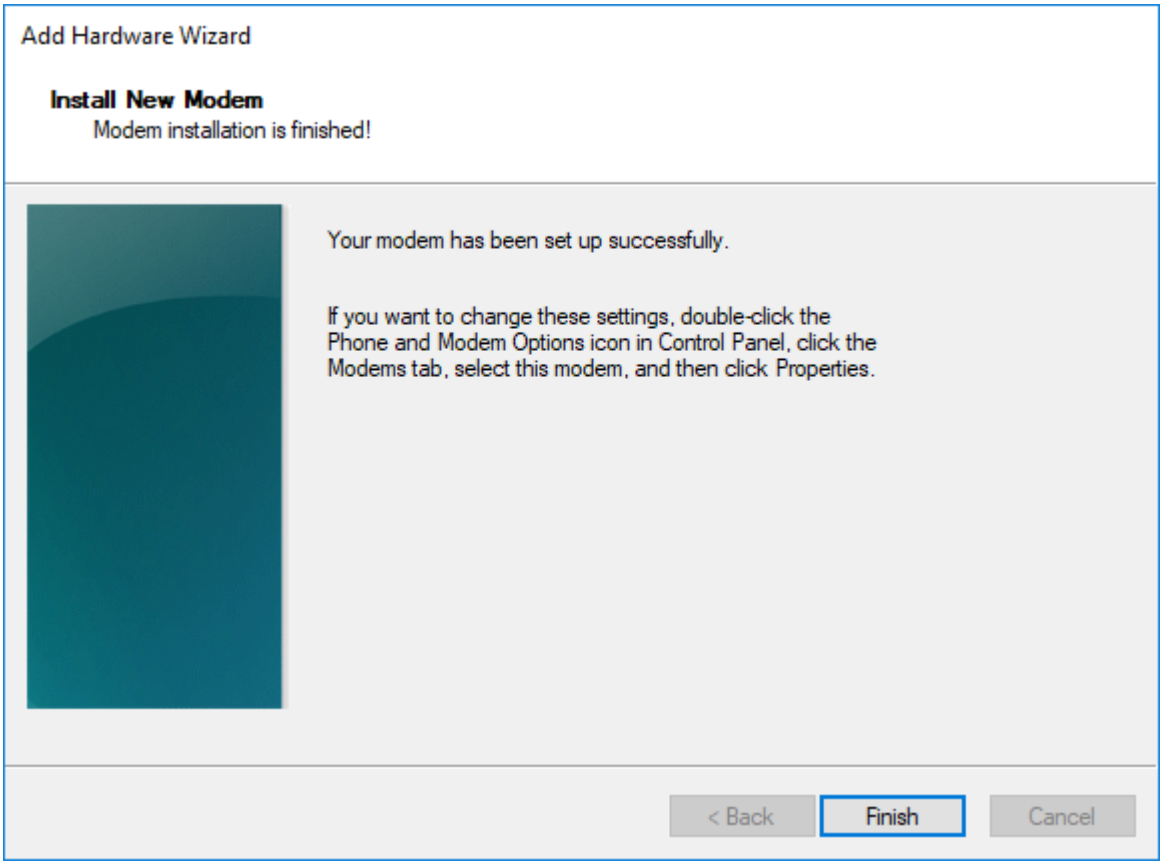


Select the COM port that is connected to the radio.

Click 'Next'.

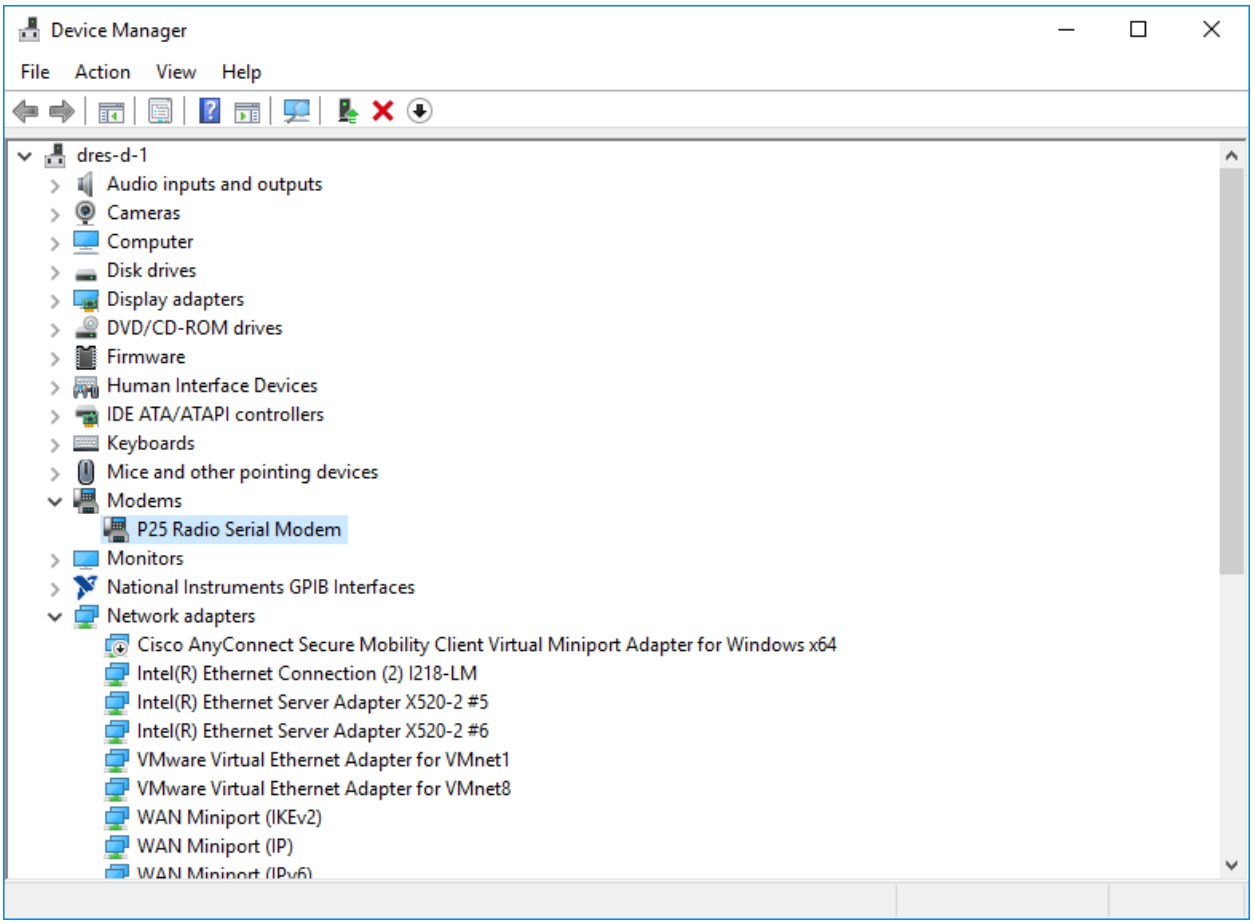


Click ‘Finish’.



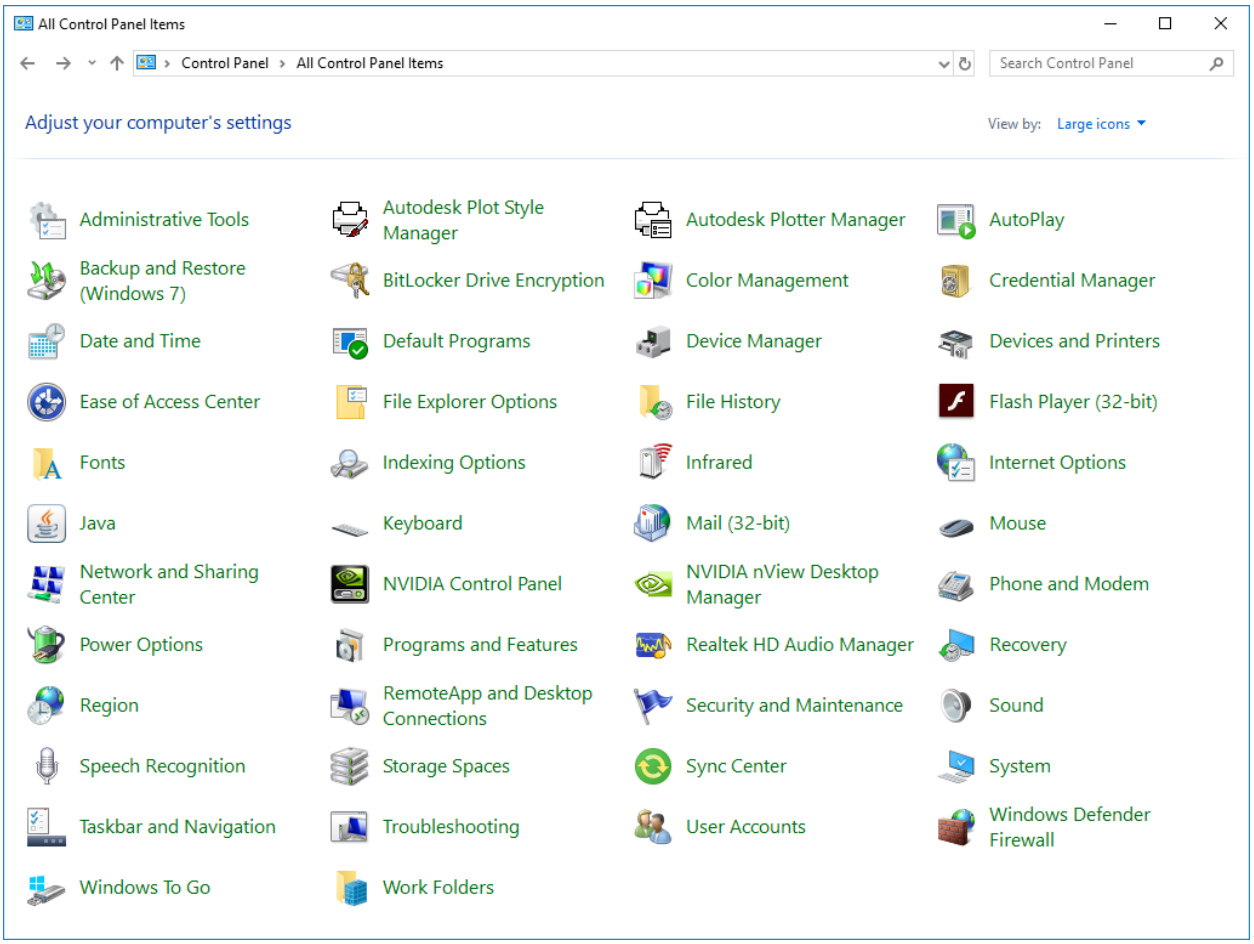
Verify that the 'P25 Radio Serial Modem' item appears under the 'Modems' category.

You may have to select 'Action'->'Scan for hardware changes' for the modem to appear.



Open the Control Panel.

Click ‘Phone and Modem’.

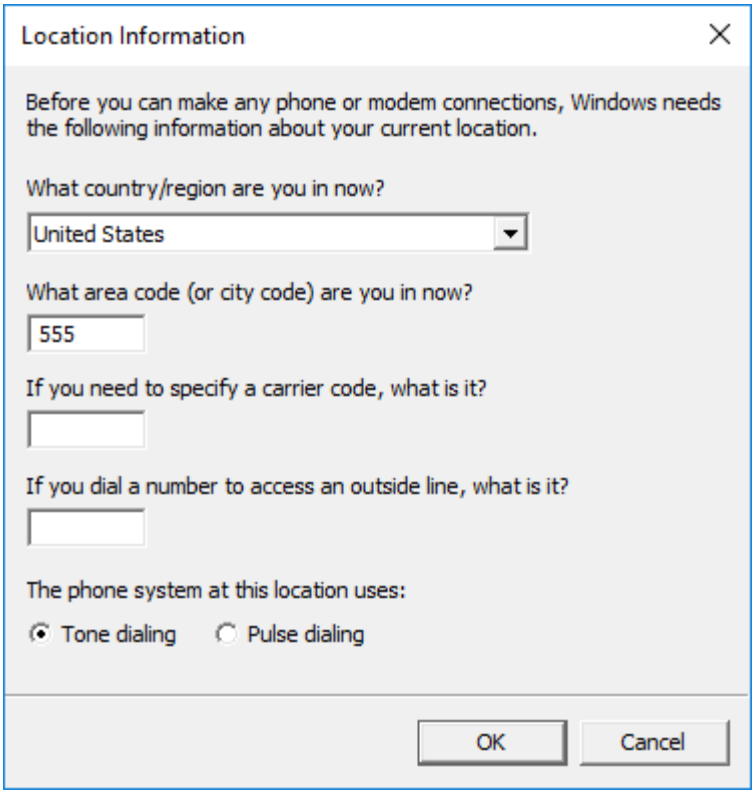


If this window appears:

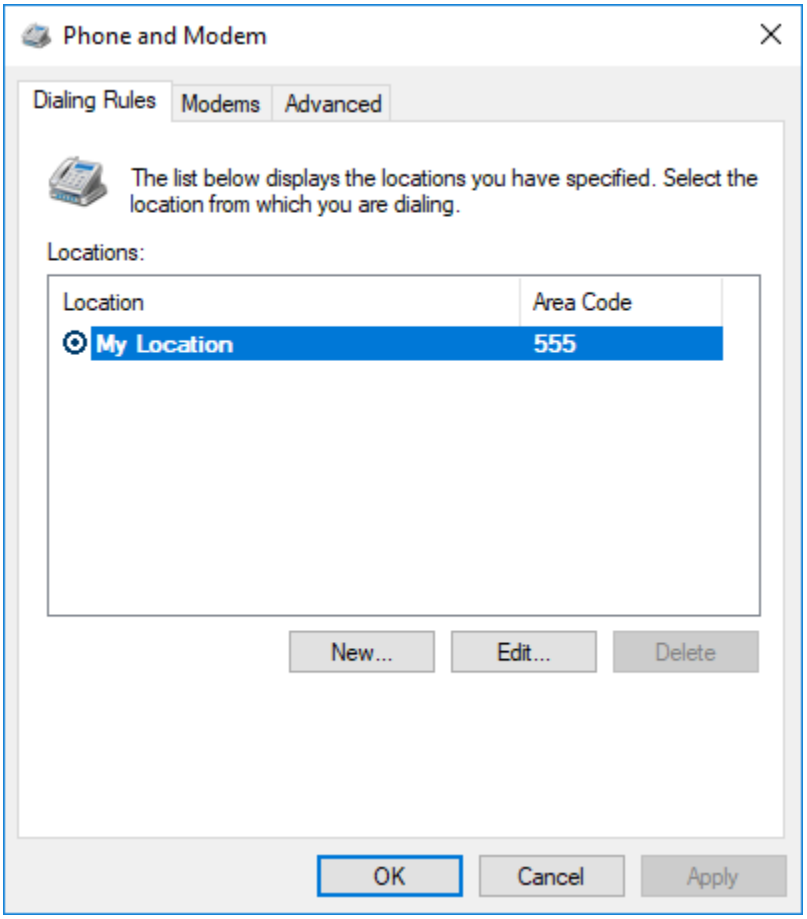
Select ‘United States’ from the ‘What country/region are you in now?’ combo box.

Enter ‘555’ into the ‘What area code (or city code) are you in now?’ text box.

Click ‘OK’.

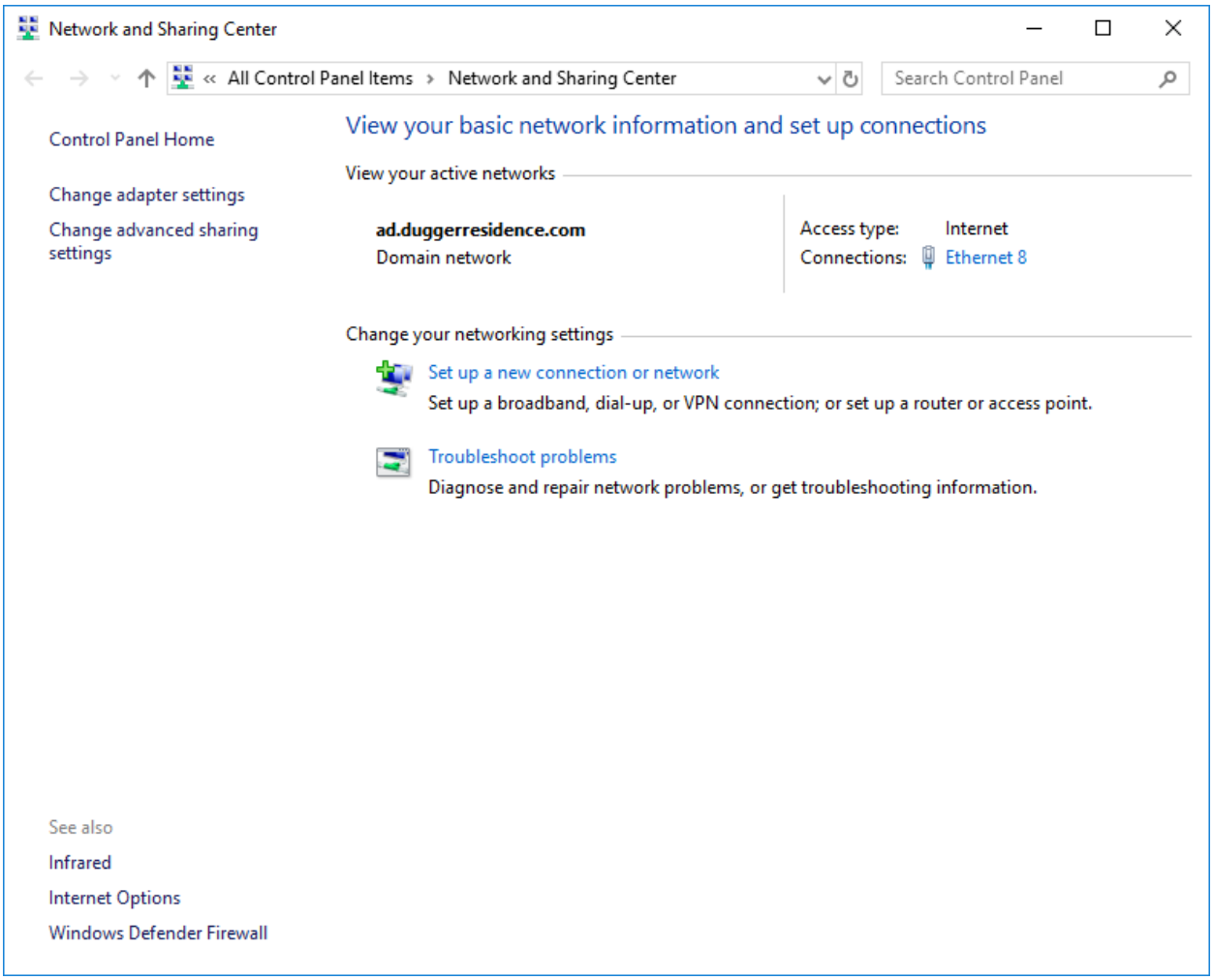


Click ‘OK’.


Phone and Modem
Dialing Rules
Modems
Advanced
The list below displays the locations you have specified. Select the location from which you are dialing.
Locations:
Location
Area Code
My Location
555
New...
Edit...
Delete
OK
Cancel
Apply

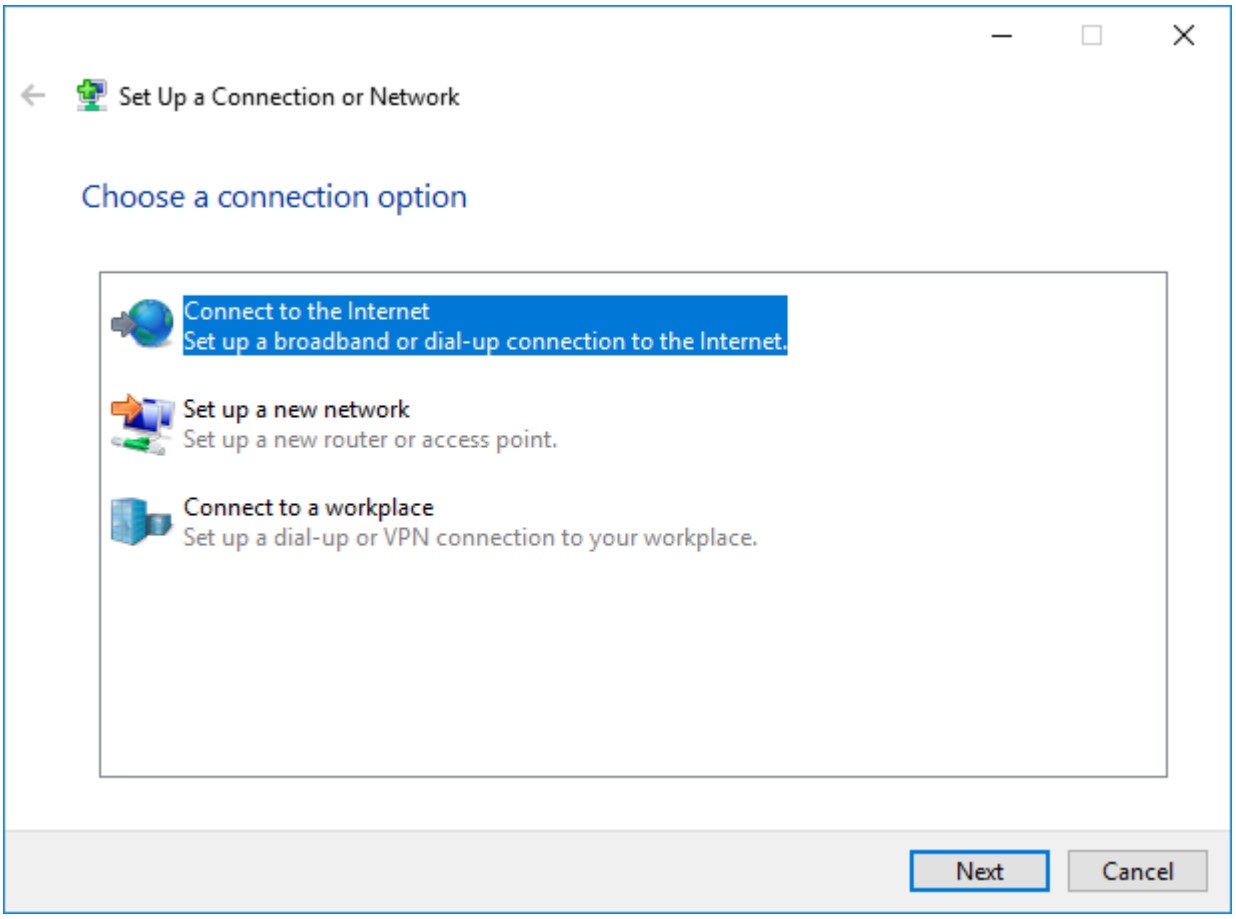
Open the Network and Sharing Center.

Click 'Set up a new connection or network'.

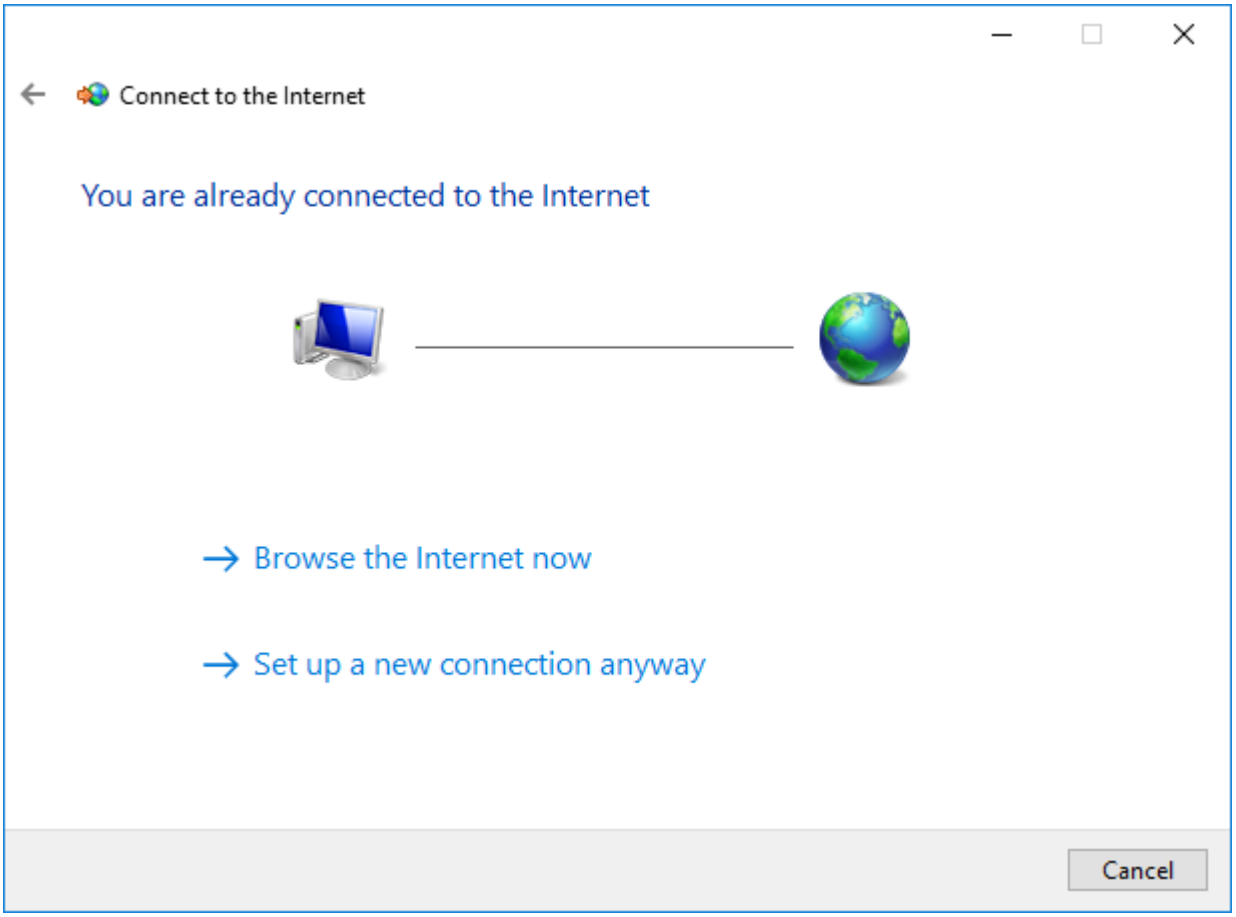


Select ‘Connect to the Internet’.

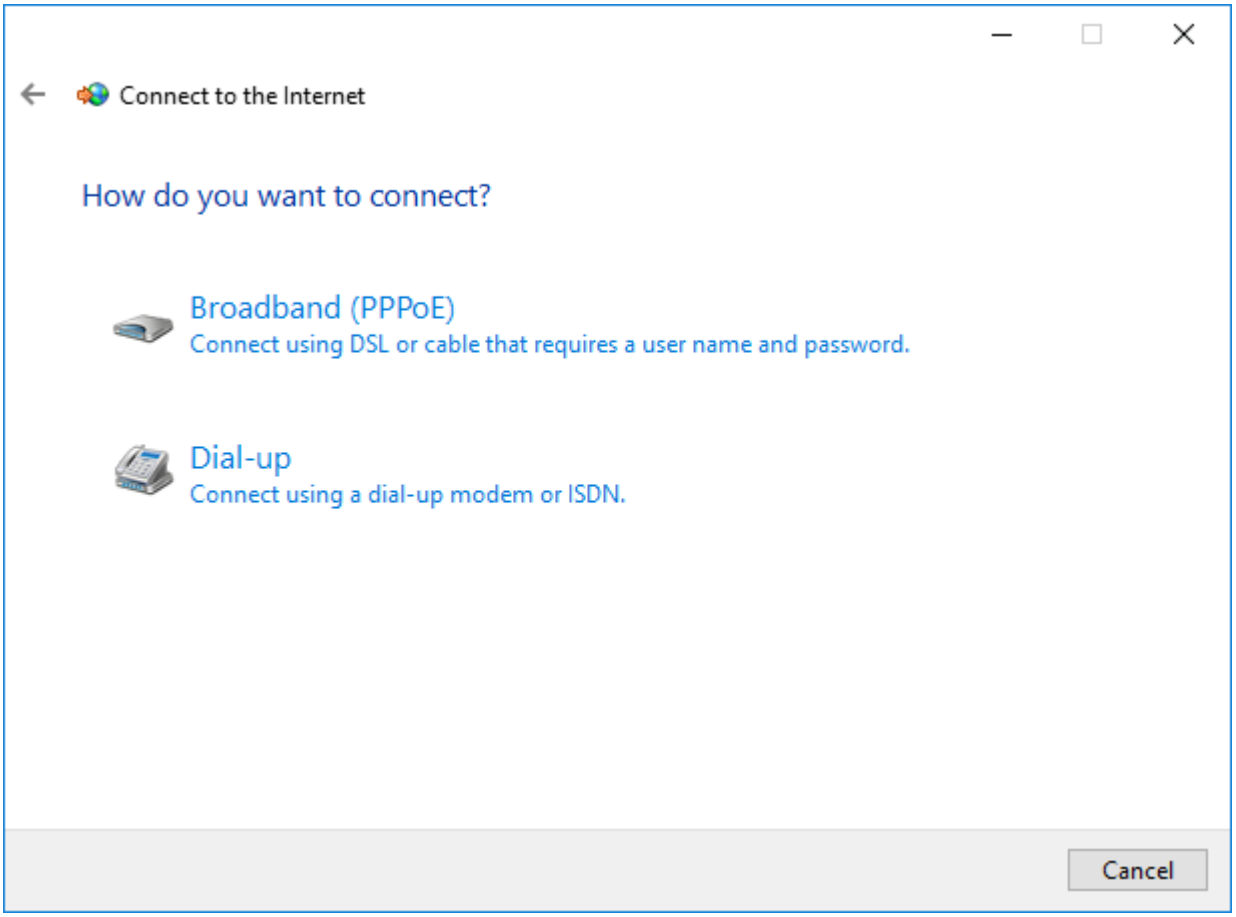
Click ‘Next’.



Click ‘Set up a new connection anyway’.



Click ‘Dial-up’.

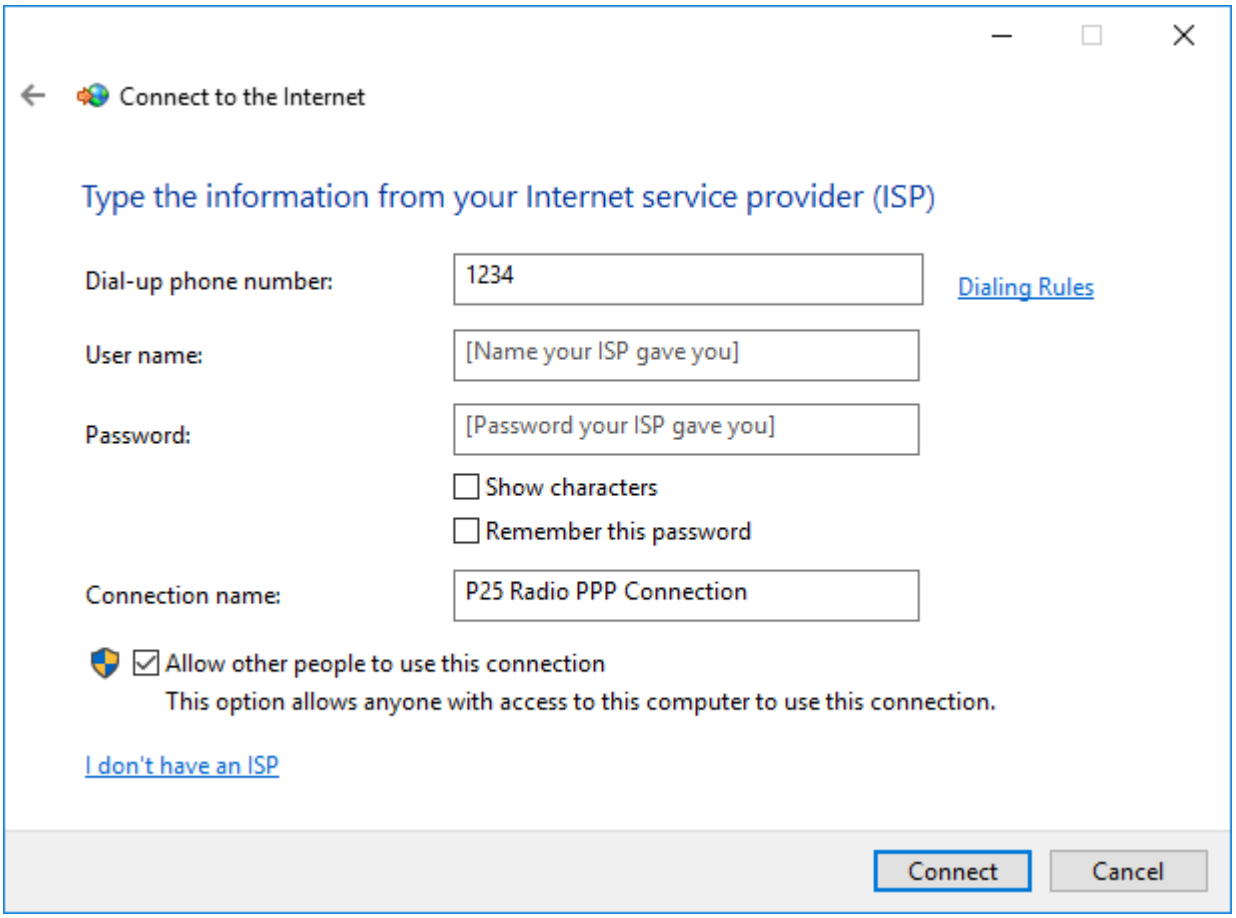


Enter ‘1234’ (or any number) into the ‘Dial-up phone number’ text box.

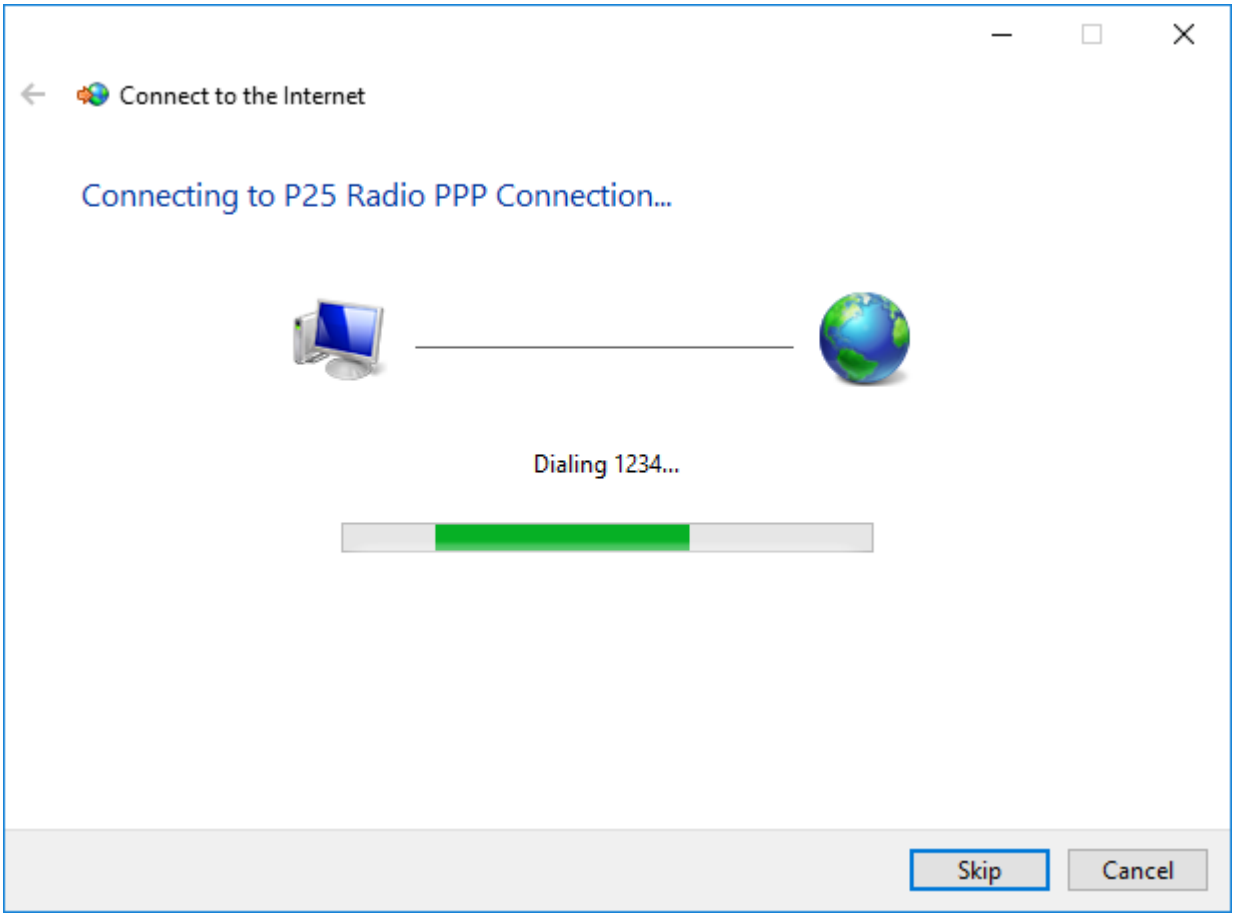
Enter ‘P25 Radio PPP Connection’ (or any name) into the ‘Connection name’ text box.

Select ‘Allow other people to use this connection.

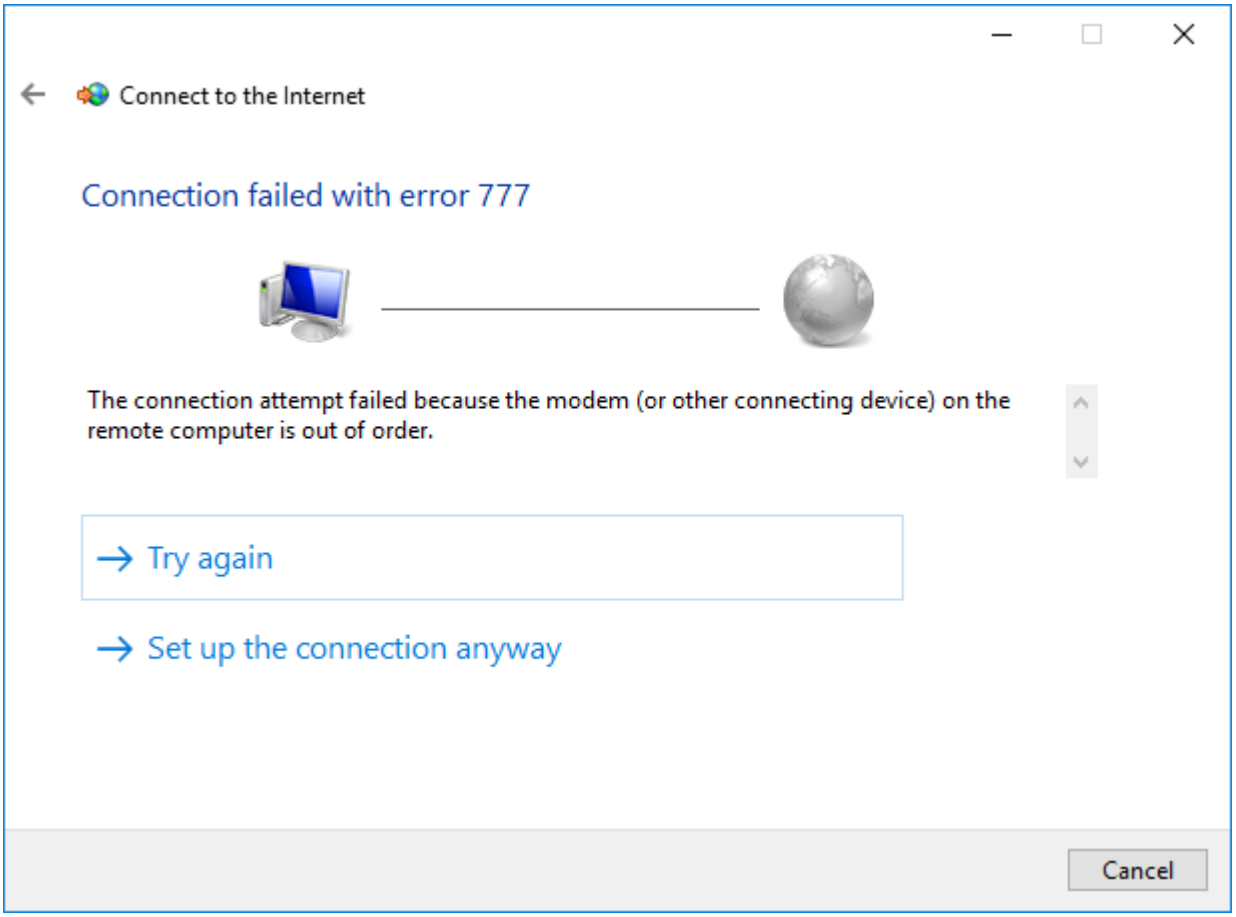
Click ‘Connect’.



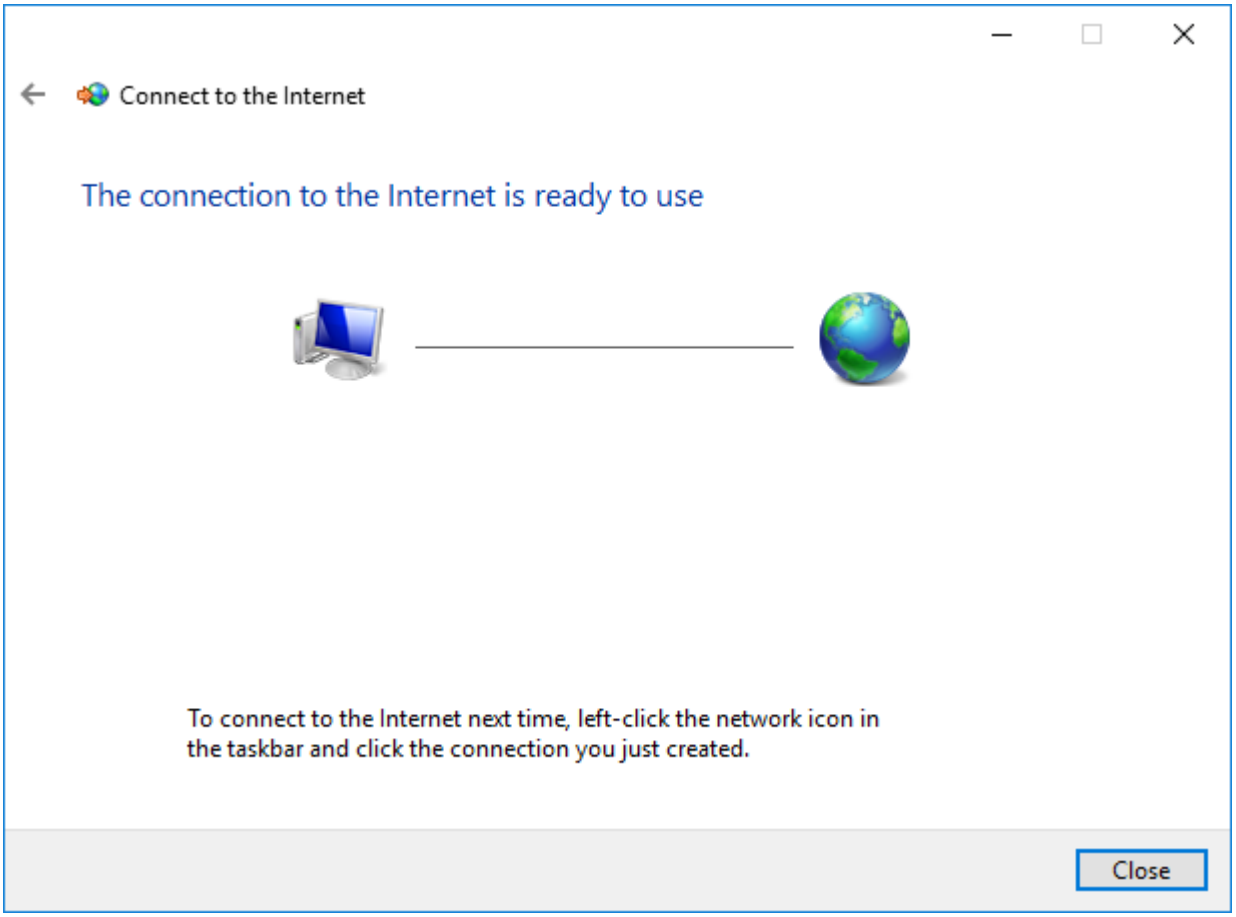
Click ‘Skip’.


Connect to the Internet
Connecting to P25 Radio PPP Connection...
Dialing 1234...
Skip
Cancel

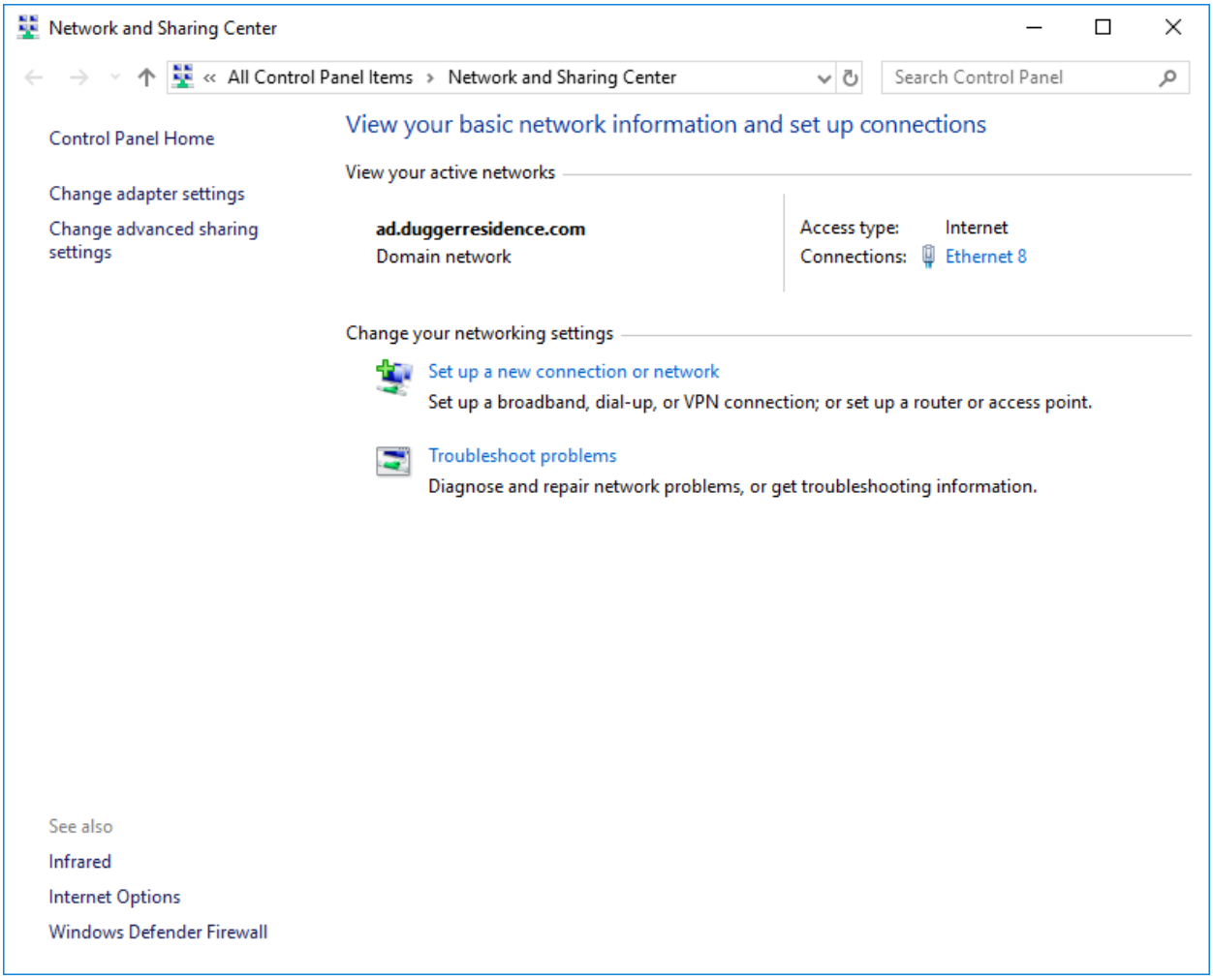
If this message appears, click 'Set up the connection anyway'.



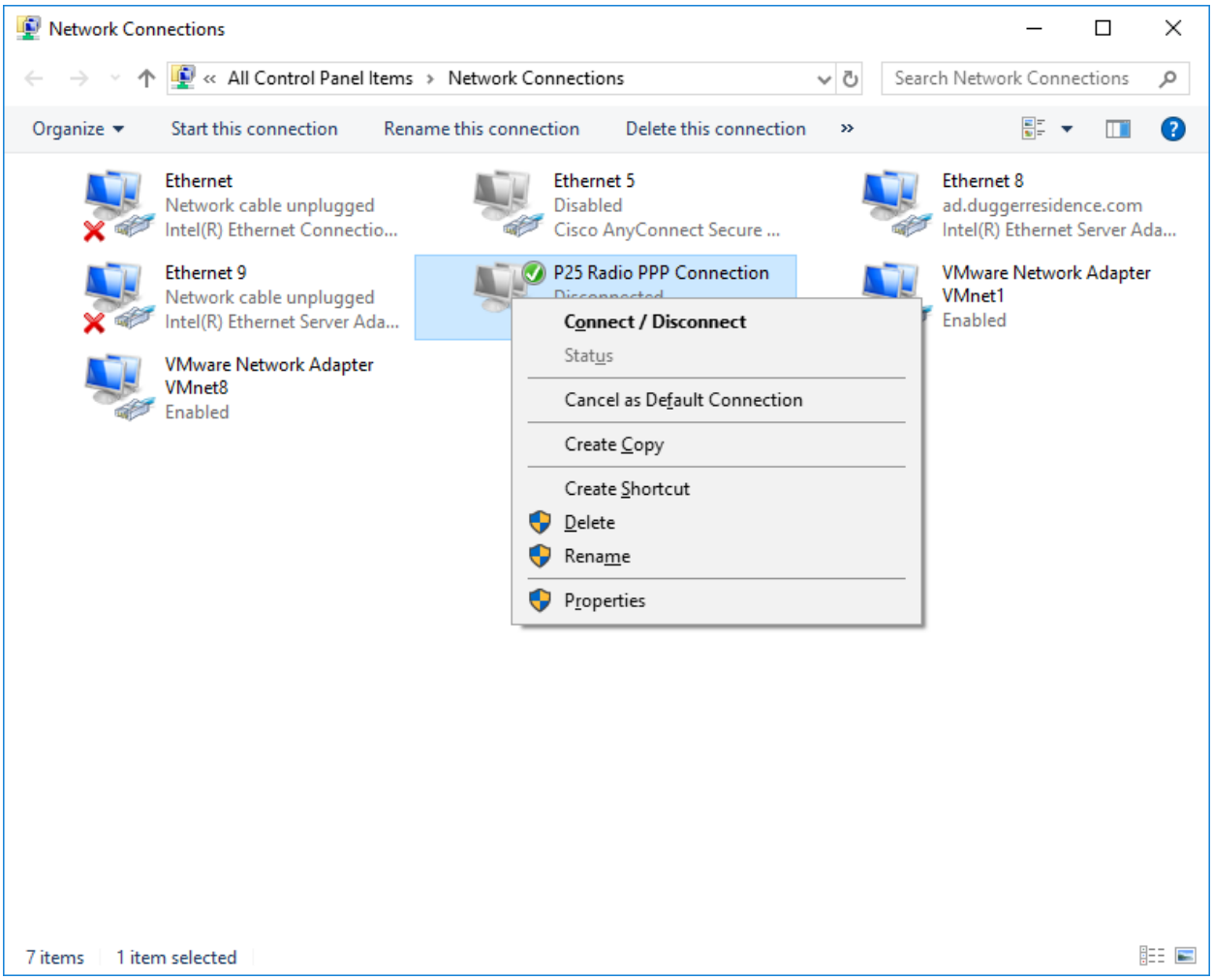
Click ‘Close’.


Connect to the Internet
The connection to the Internet is ready to use
To connect to the Internet next time, left-click the network icon in the taskbar and click the connection you just created.
Close

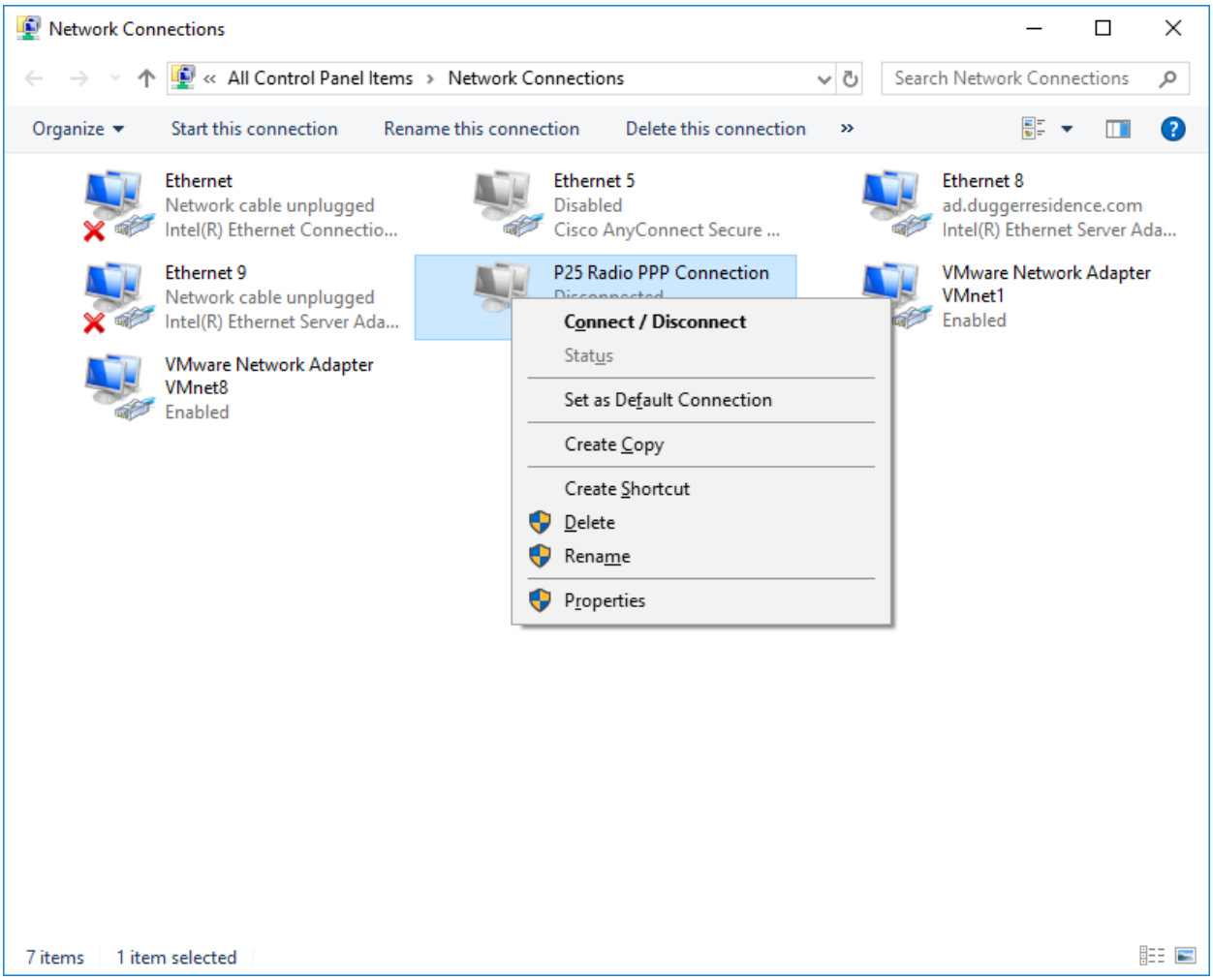
Click 'Change Adapter Settings'



Right click 'P25 Radio PPP Connection' (or whatever you named the dial-up connection), and click 'Cancel as Default Connection'.



Right click 'P25 Radio PPP Connection' (or whatever you named the dial-up connection), and click 'Properties'.

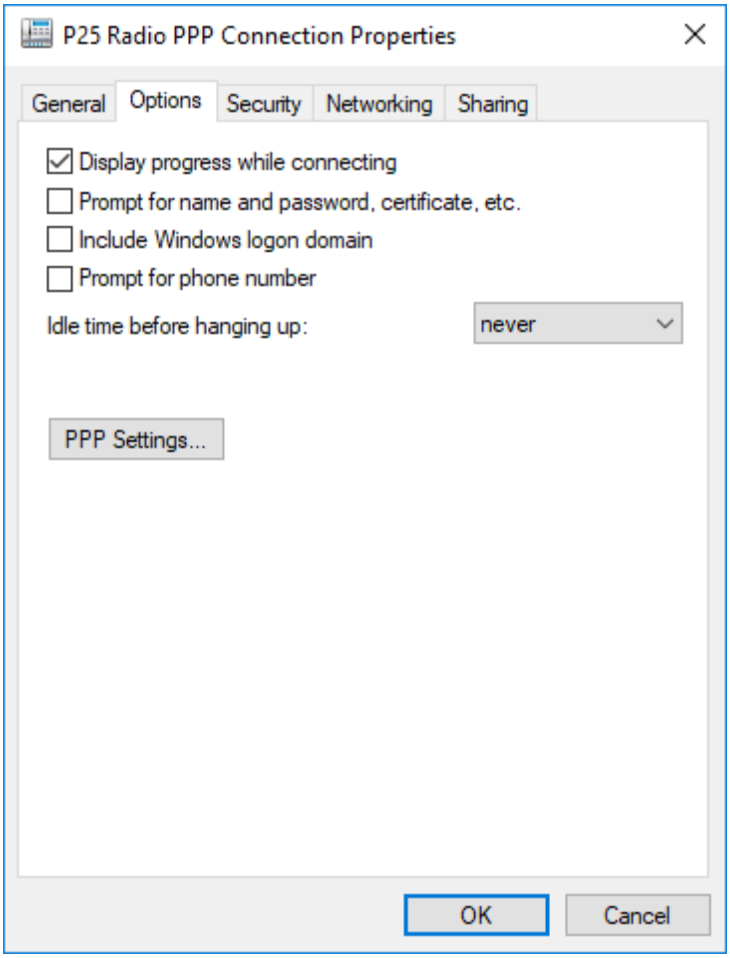


Click ‘Options’

Uncheck ‘Prompt for name and password, certificate, etc’.

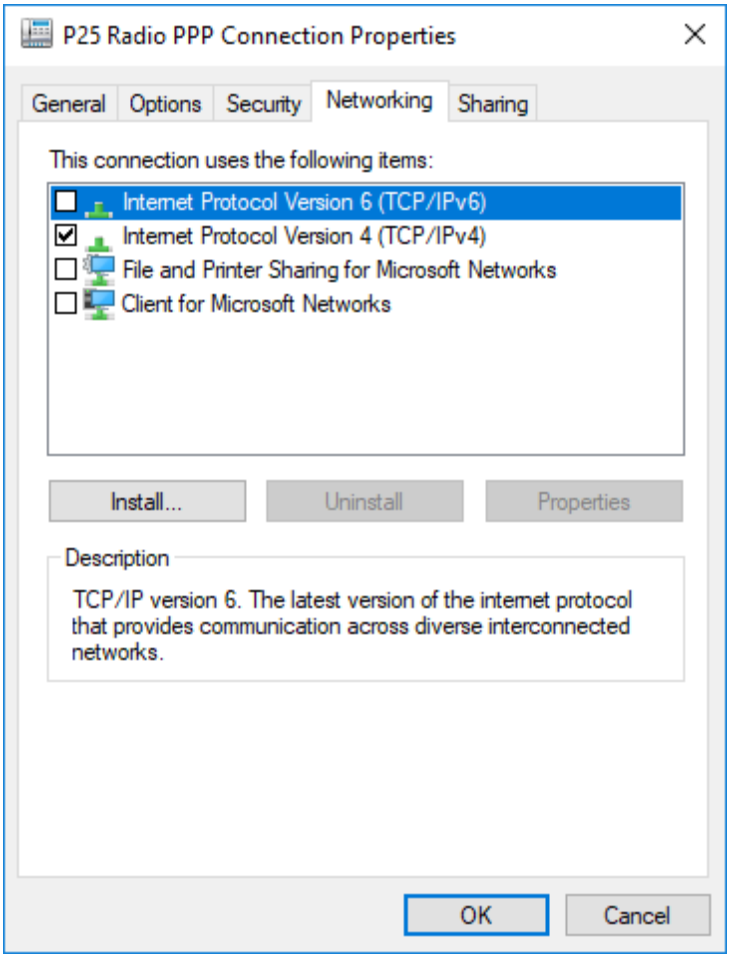
Uncheck ‘Prompt for phone number’.

Select ‘never’ from the ‘Idle time before hanging up’ combo box.



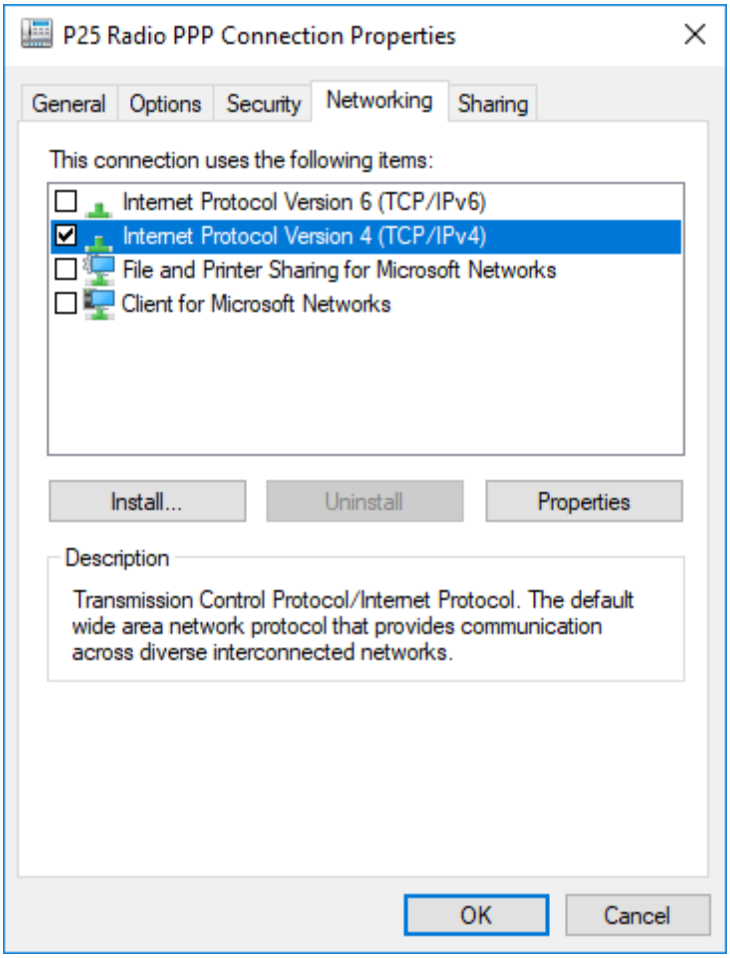
Click ‘Networking’

Uncheck ‘Internet Protocol Version 6 (TCP/IPv6)’.

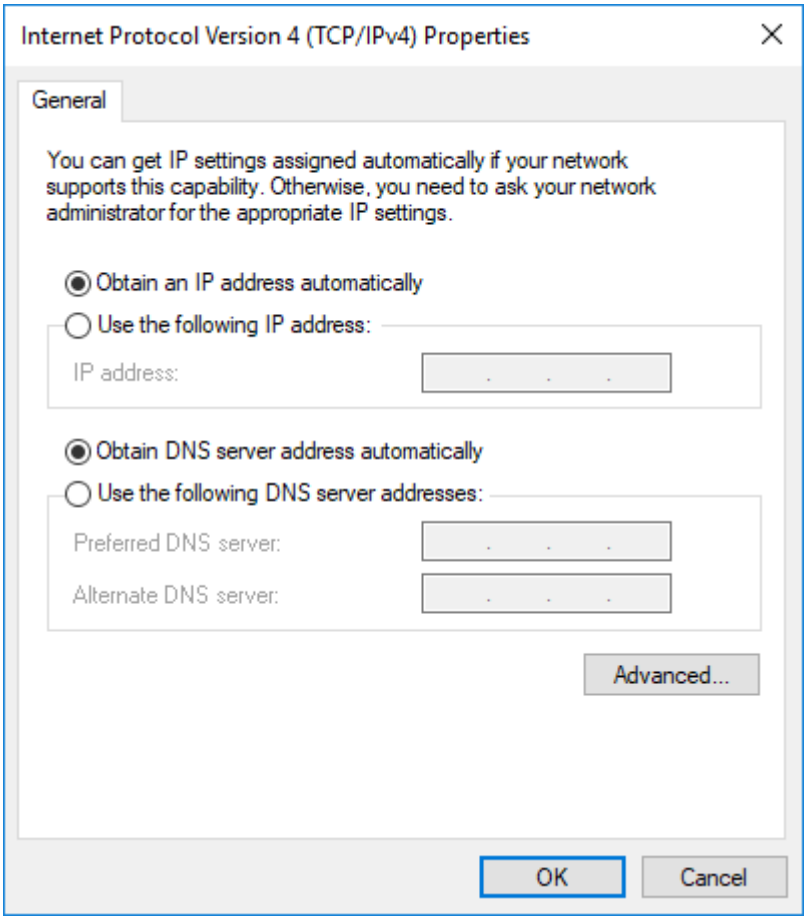


Click ‘Internet Protocol Version 4 (TCP/IPv4)’.

Click ‘Properties’.

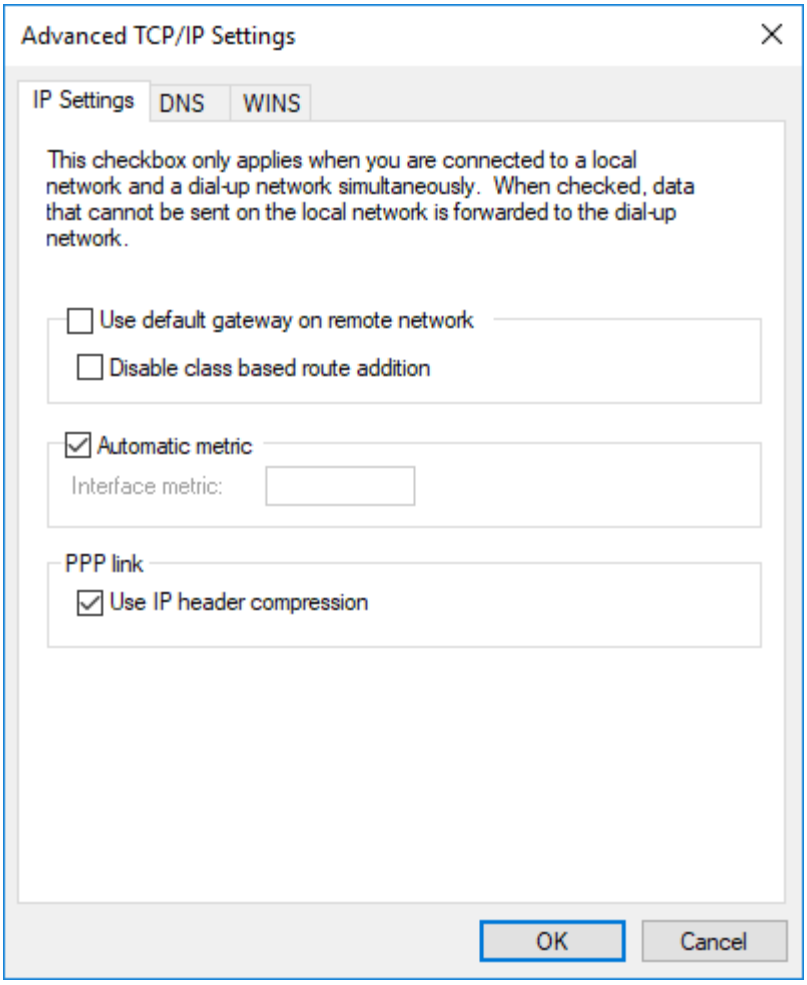


Click ‘Advanced’.

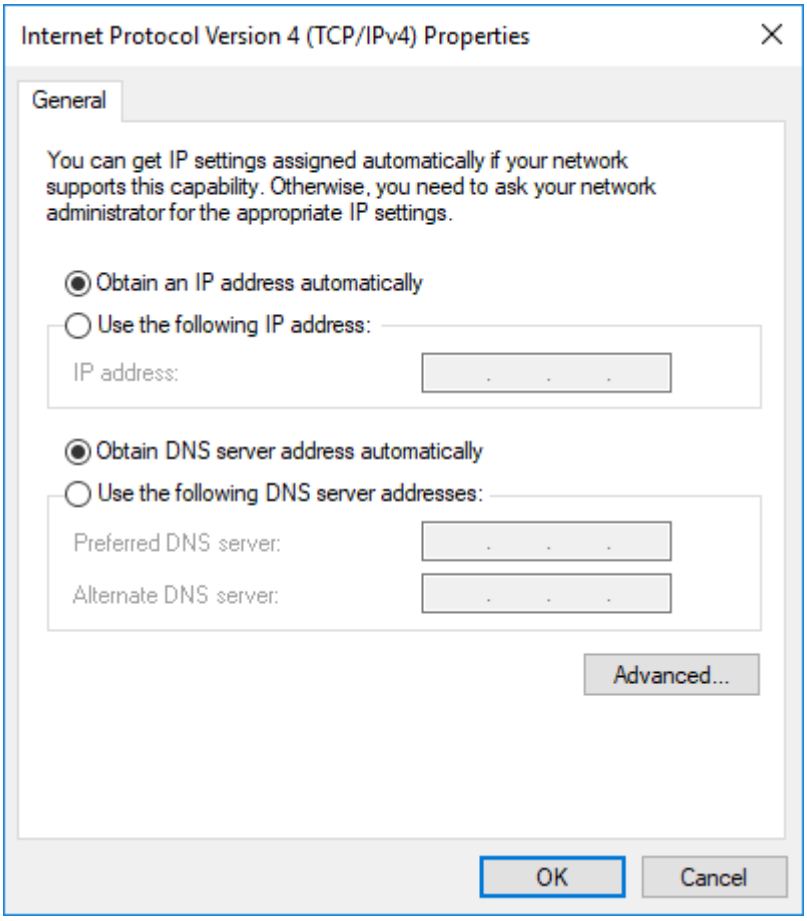


Uncheck ‘Use default gateway on remote network’.

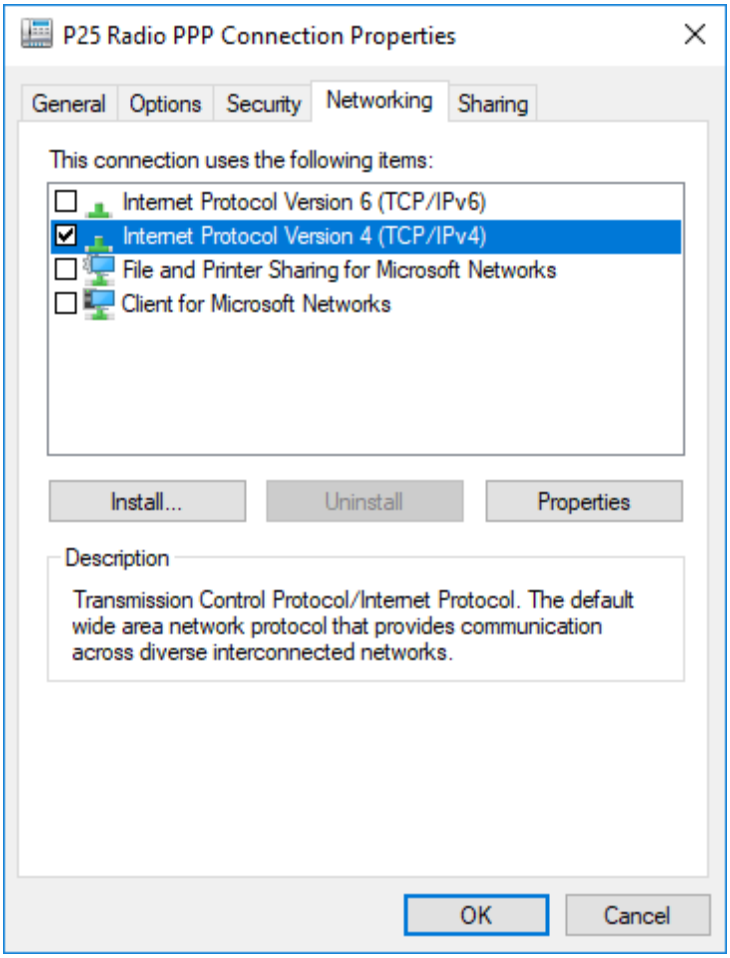
Click ‘OK’.



Click ‘OK’.


Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Advanced...
OK
Cancel

Click ‘OK’.

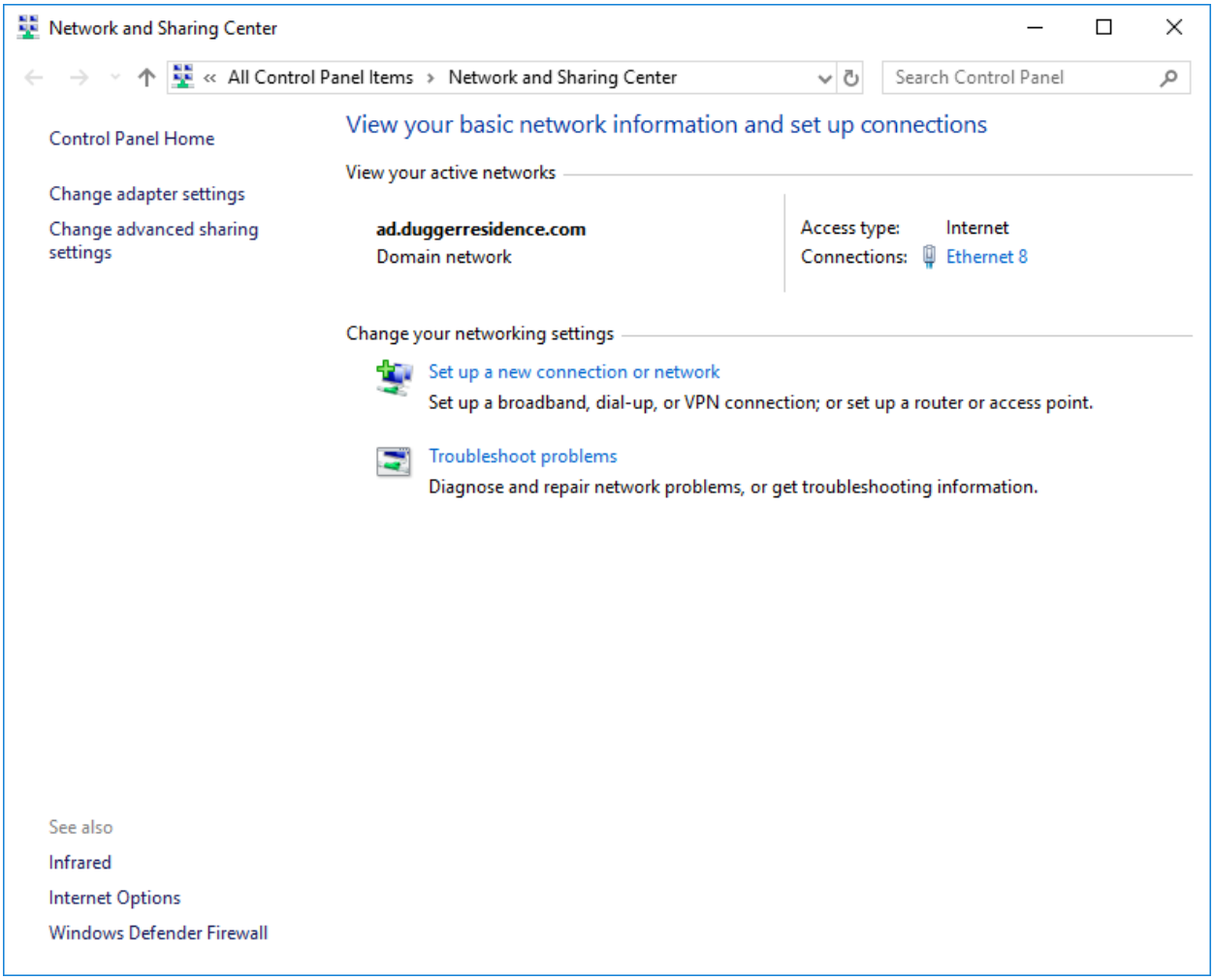

P25 Radio PPP Connection Properties
General
Options
Security
Networking
Sharing
This connection uses the following items:
Internet Protocol Version 6 (TCP/IPv6)
Internet Protocol Version 4 (TCP/IPv4)
File and Printer Sharing for Microsoft Networks
Client for Microsoft Networks
Install...
Uninstall
Properties
Description
Transmission Control Protocol/Internet Protocol. The default wide area network protocol that provides communication across diverse interconnected networks.
OK
Cancel

Connection

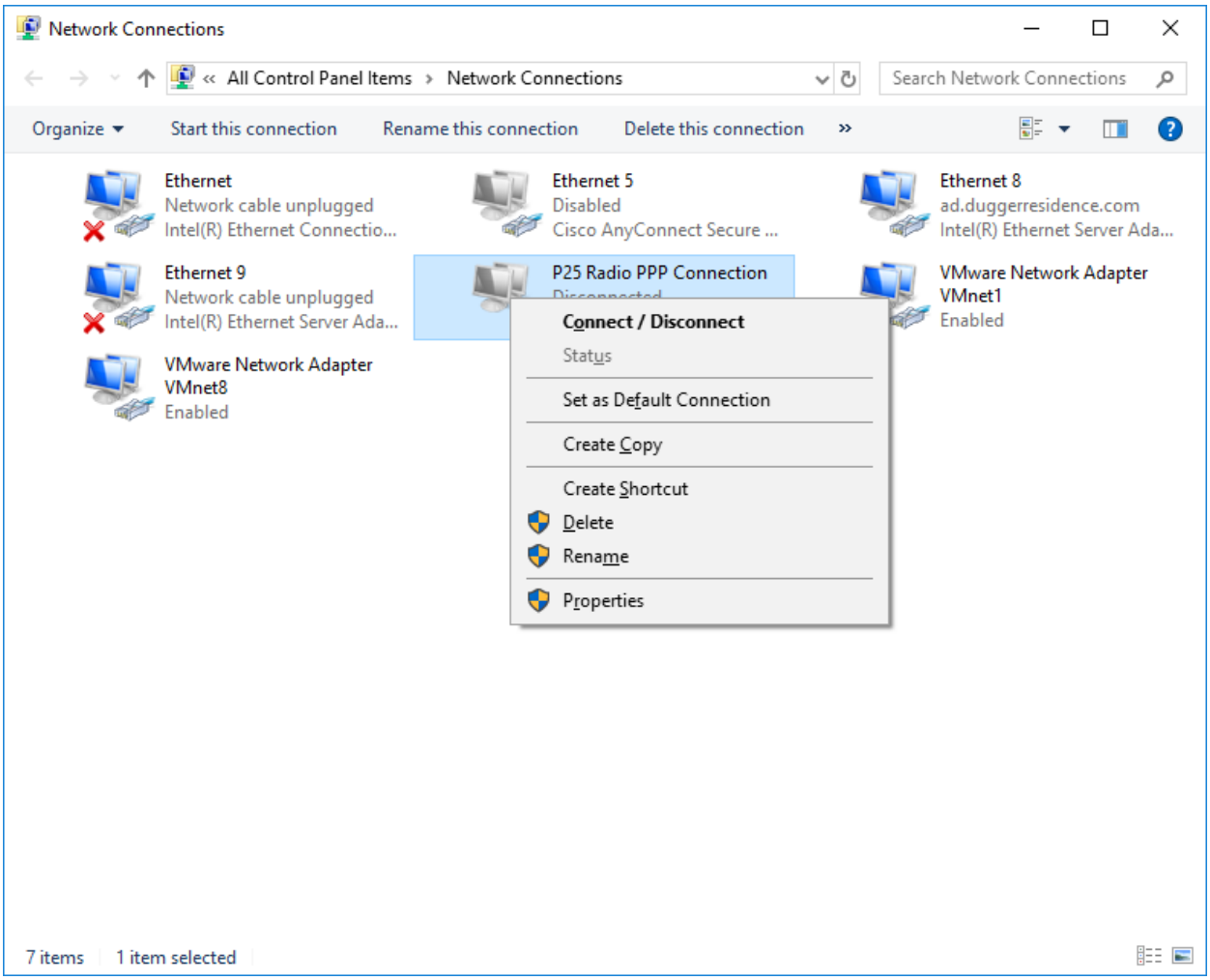
Connect the radio to the serial port.

Open the Network and Sharing Center.

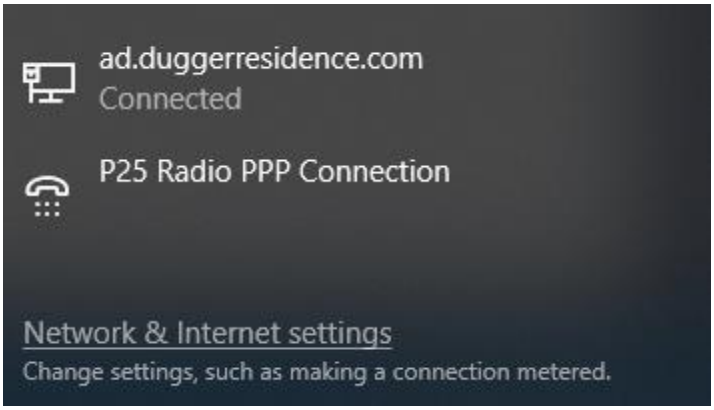
Click 'Change Adapter Settings'.



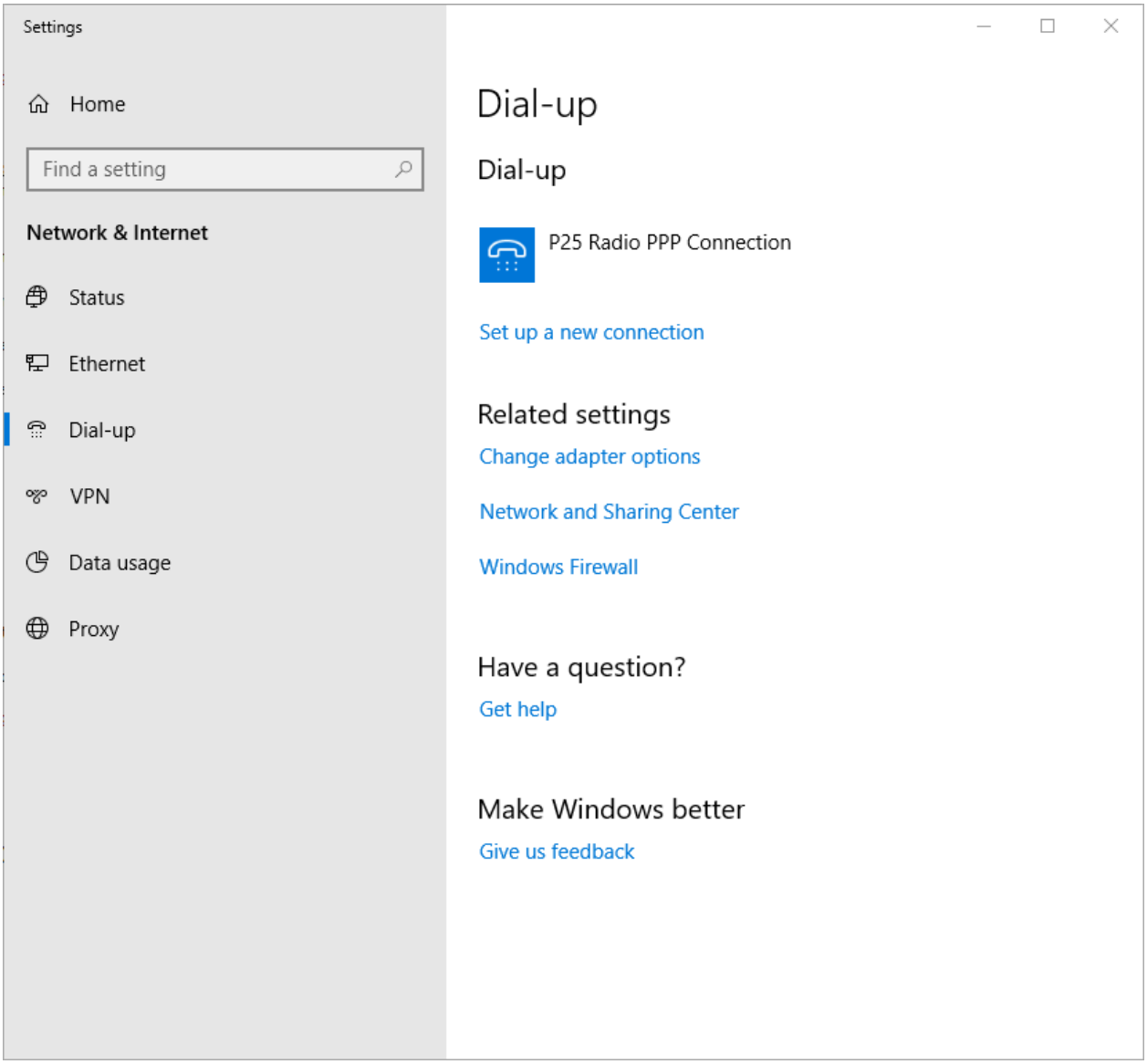
Right click 'P25 Radio PPP Connection' (or whatever you named the dial-up connection), and click 'Connect / Disconnect.



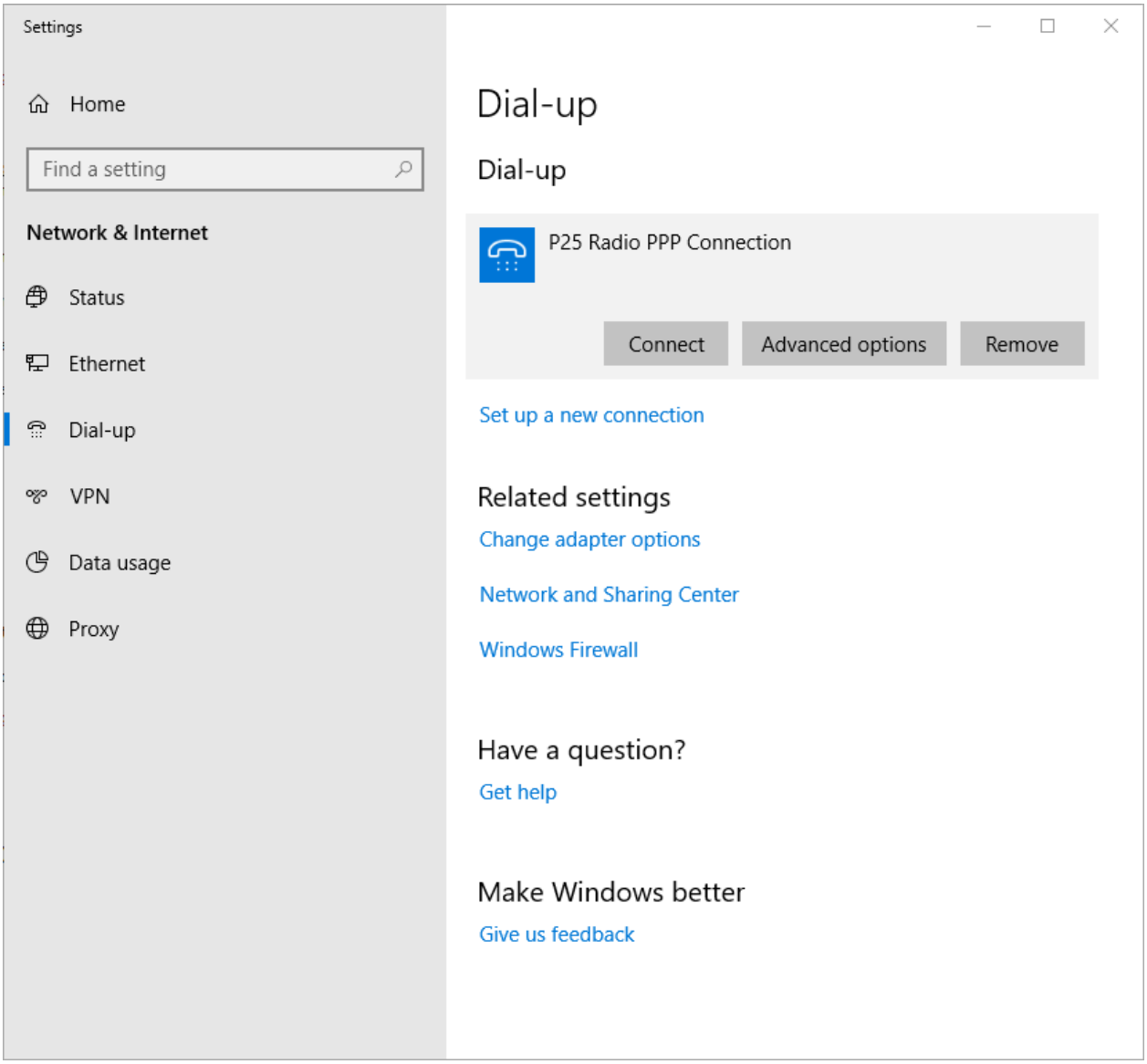
Click ‘P25 Radio PPP Connection’ (or whatever you named the dial-up connection).



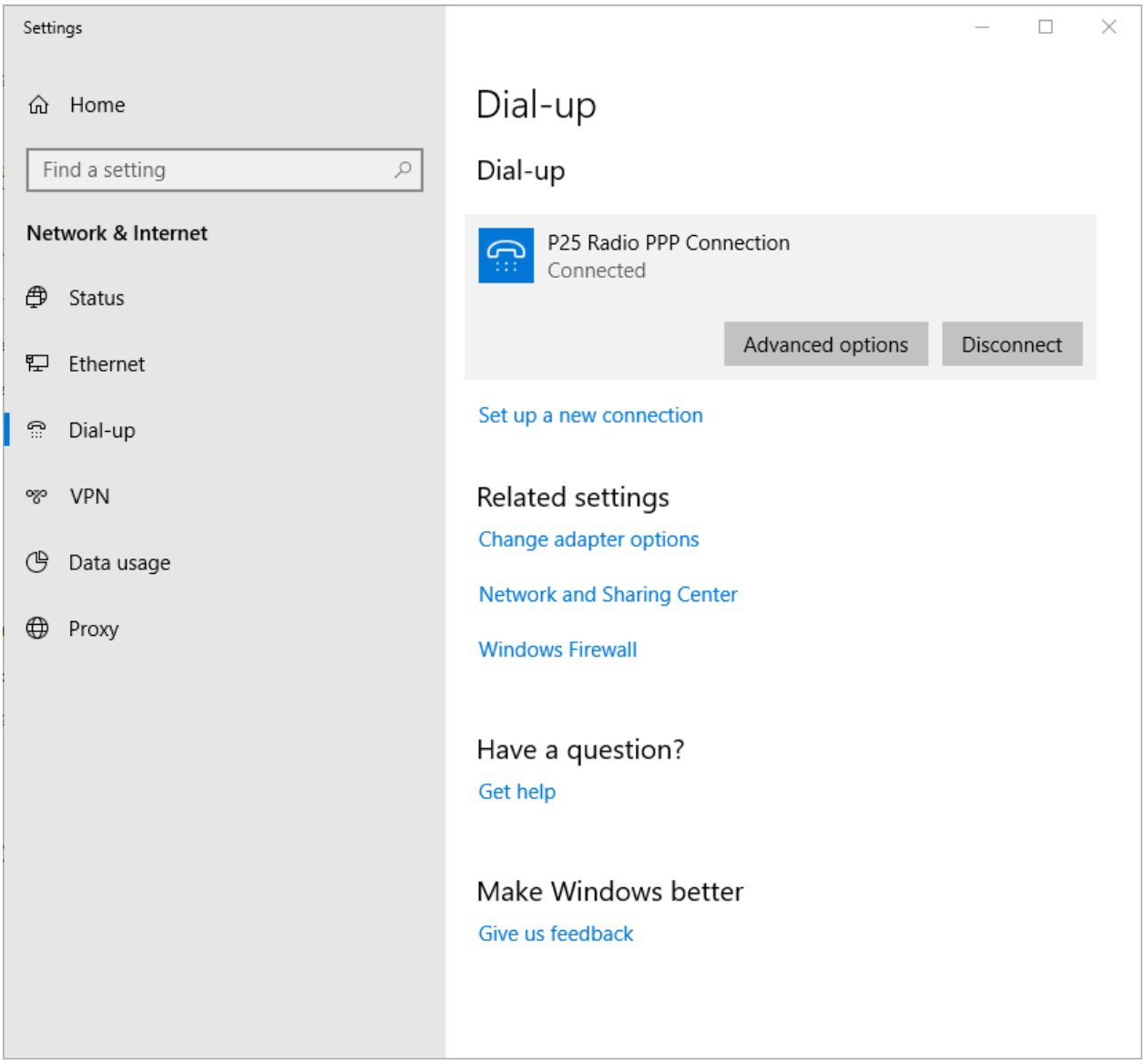
Click ‘P25 Radio PPP Connection’ (or whatever you named the dial-up connection).



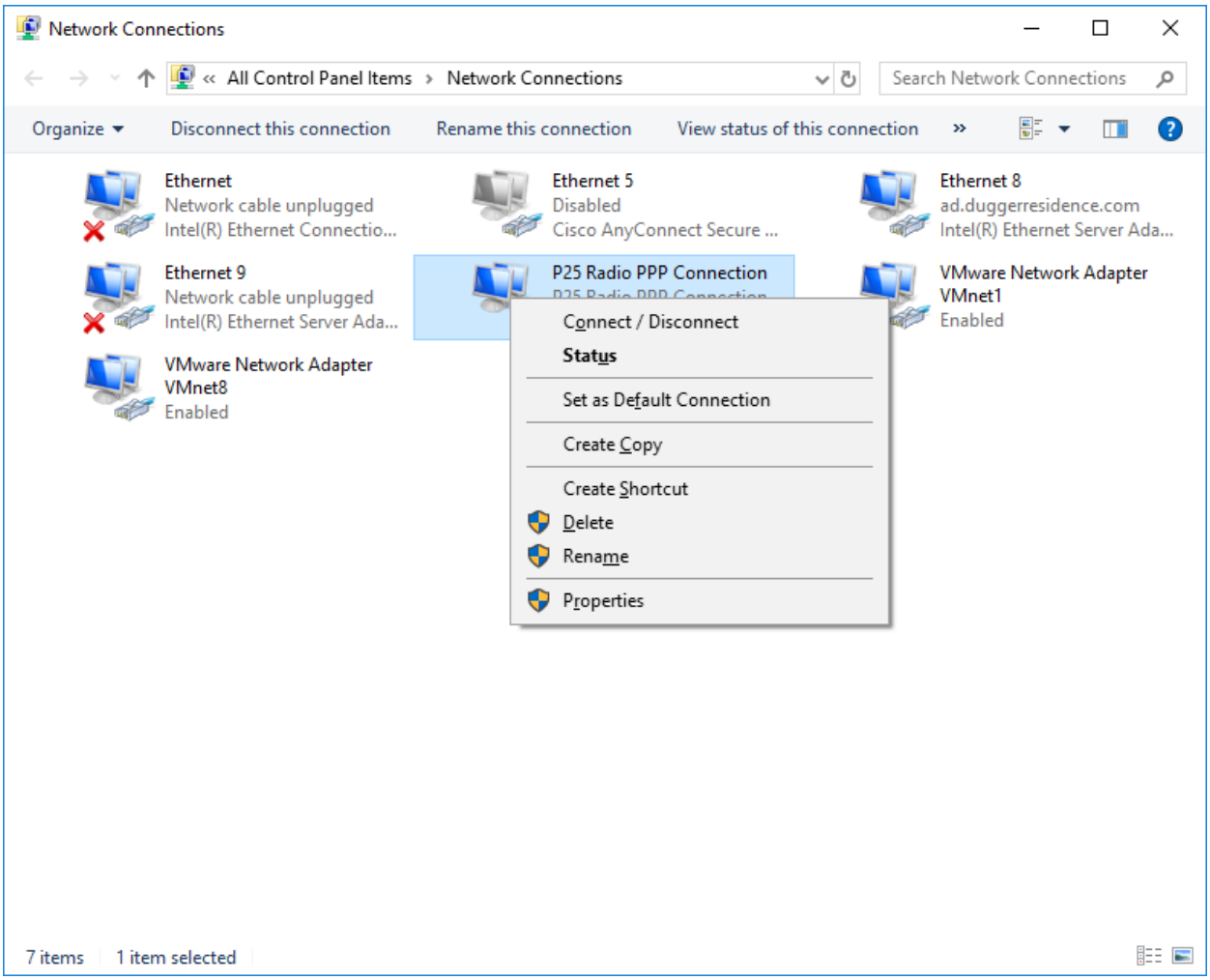
Click ‘Connect’.



Verify that the 'P25 Radio Serial Modem' (or whatever you named the dial-up connection) shows 'Connected'.



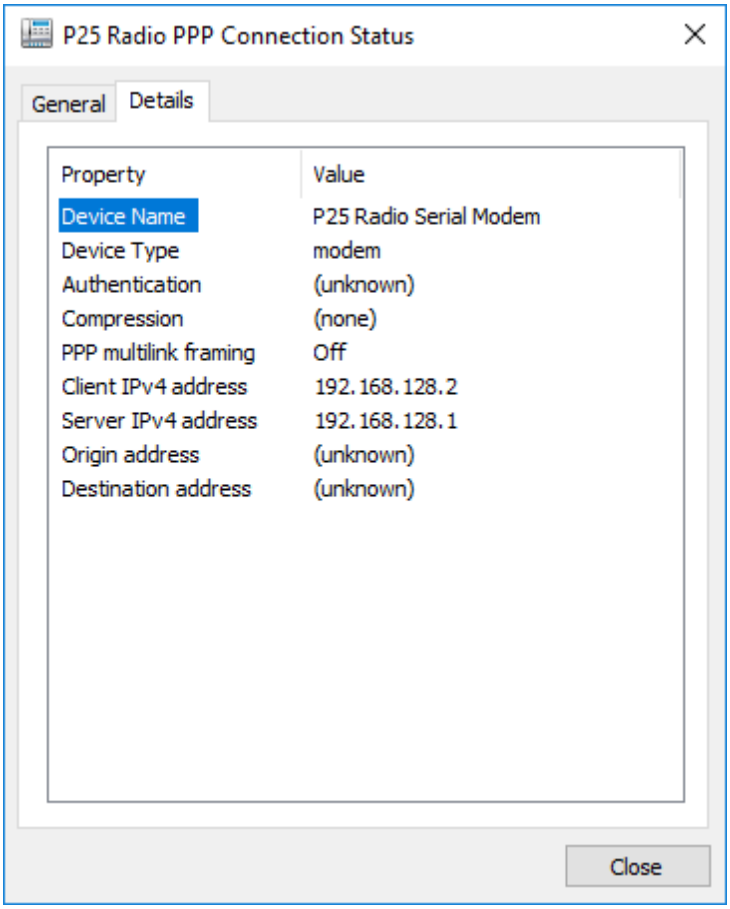
Right click 'P25 Radio PPP Connection' (or whatever you named the dial-up connection), and click 'Status.



Click ‘Details’.

The radio IP address is what appears in the ‘Server IPv4 address’ field.

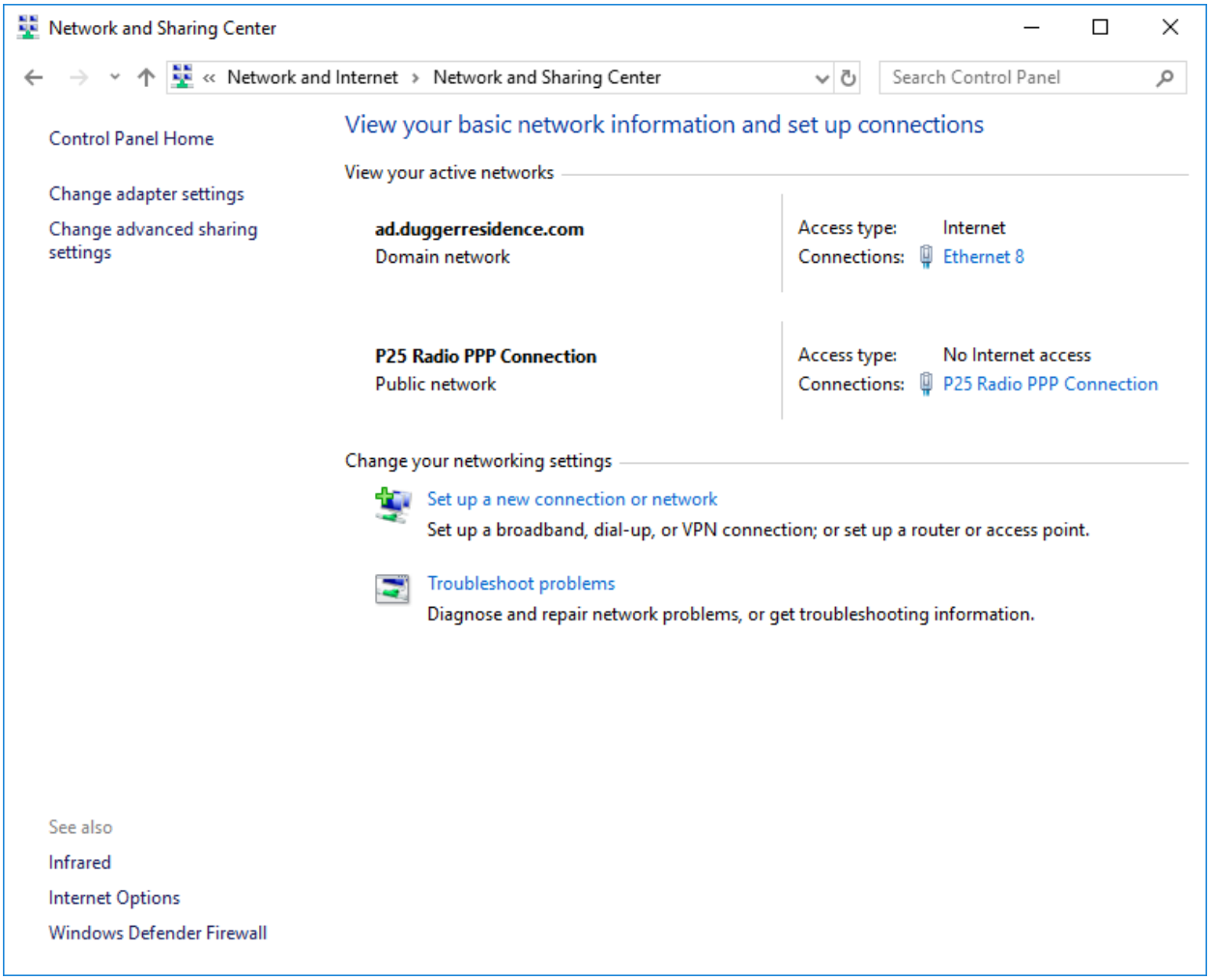
Click ‘Close’.



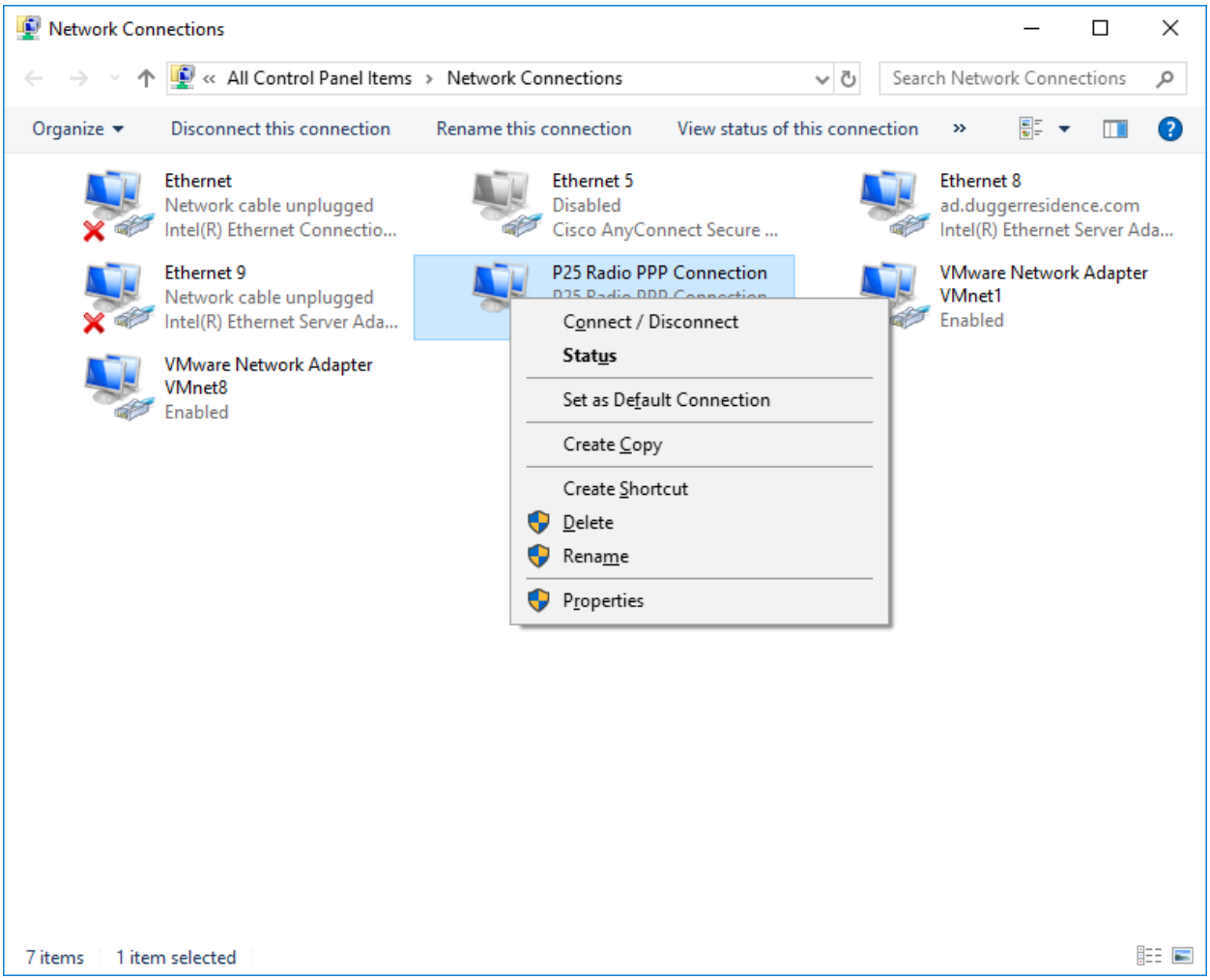
Disconnection

Open the Network and Sharing Center.

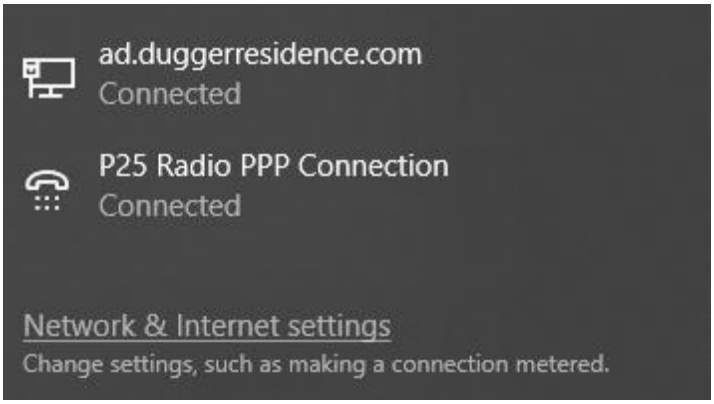
Click 'Change Adapter Settings'.



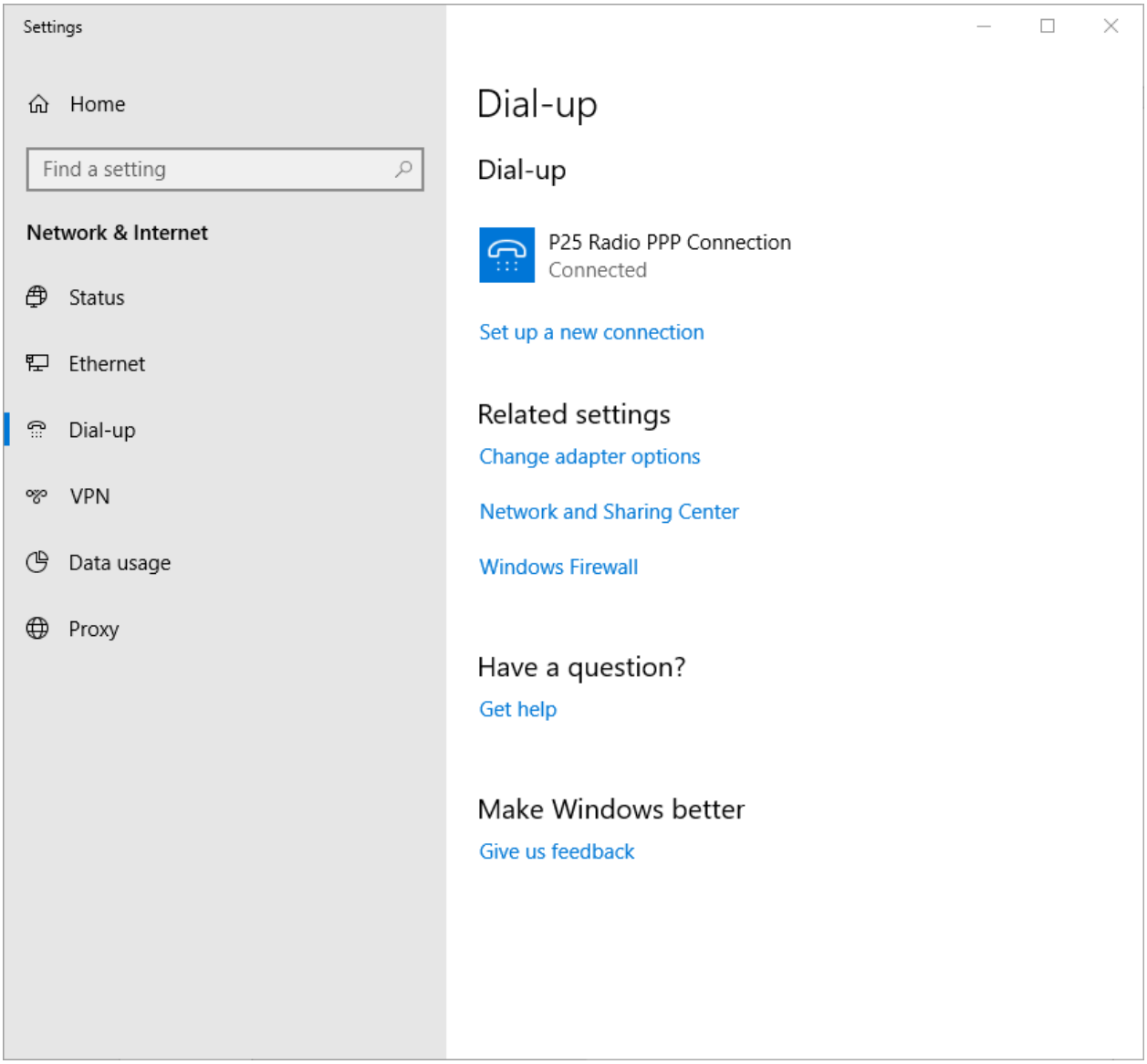
Right click ‘P25 Radio PPP Connection’ (or whatever you named the dial-up connection), and click ‘Connect / Disconnect.



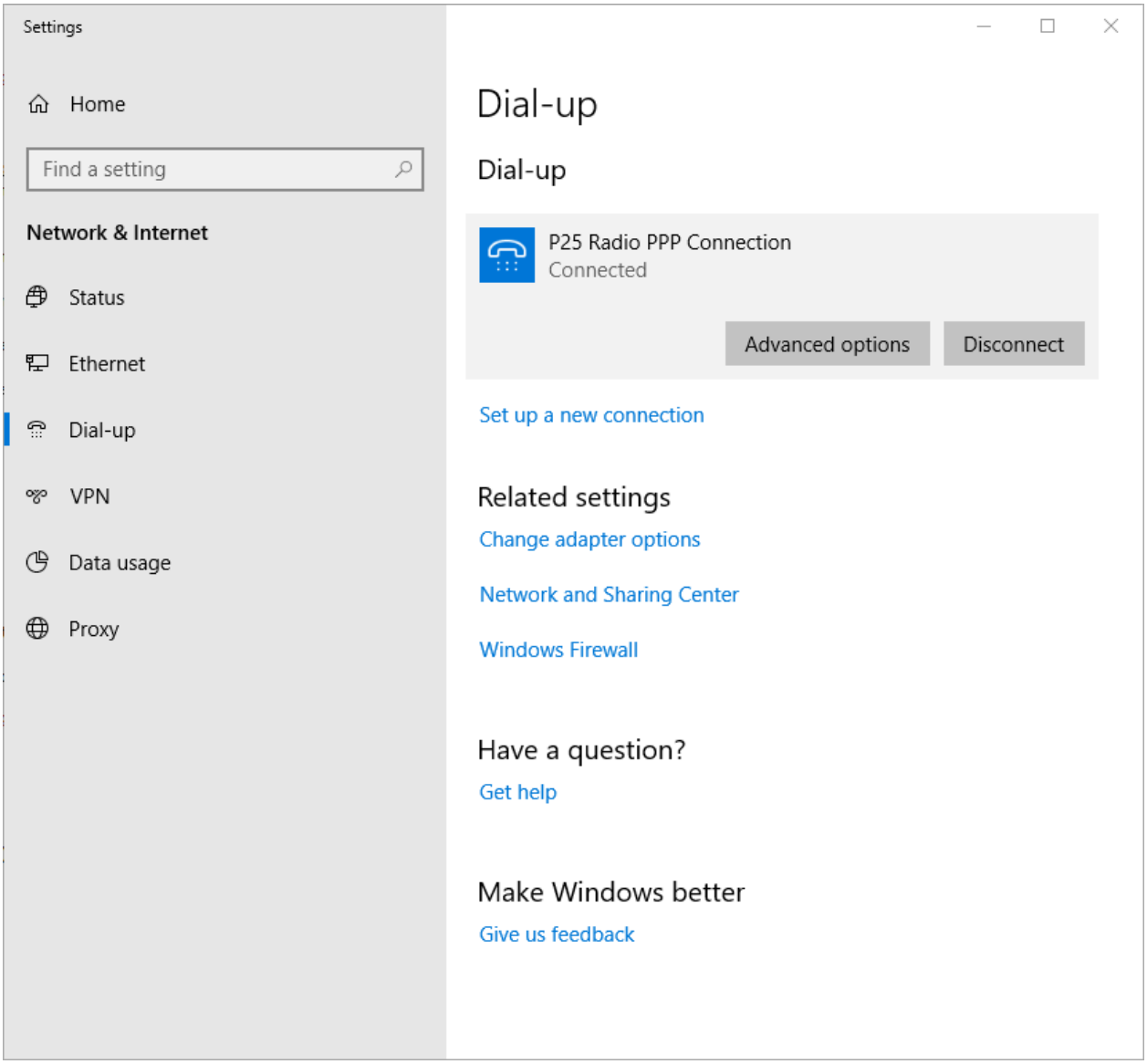
Click ‘P25 Radio PPP Connection’ (or whatever you named the dial-up connection).



Click ‘P25 Radio PPP Connection’ (or whatever you named the dial-up connection).

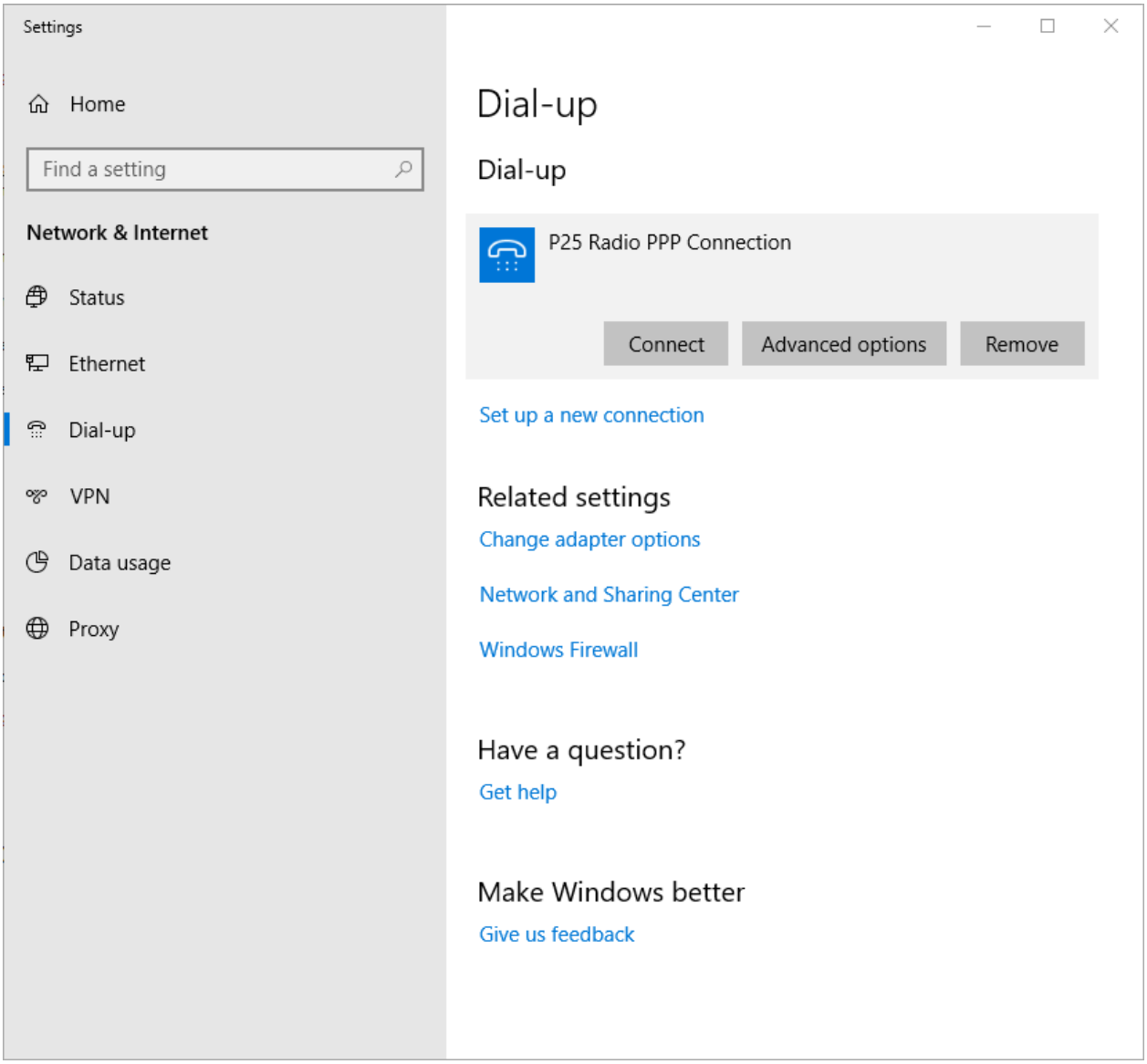


Click ‘Disconnect.



Verify that the 'P25 Radio PPP Connection' (or whatever you named the dial-up connection) does not show the 'Connected' status.

Disconnect the radio from the serial port.



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Manual Revision History

Revision #	Date	Description
1	2019-01-17	Initial Release for 1.0.0.0